

The Shetland Pony

BY

CHARLES AND ANNE DOUGLAS

With an Appendix on
The Making of the Shetland Pony

By J. COSSAR EWART

GRAVURO AND OTHER ILLUSTRATIONS

William Blackwood and Sons
Edinburgh and London

1913

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Transcriber's Notes

Footnote anchors have been changed from numeric to an alpha plus numeric sequence such as: [F1] [F2], to create a point of difference between footnotes and endnotes which are also numeric.

[Page 43](#)—both spellings of ‘maiseys’ and ‘klibbar’ from the writings of Hibbert of 1820’s and ‘maysies’ and ‘clibber’ from Cowies writings of 1874 have been retained.

Other changes made are noted at the [end of the book](#).



CHANGING QUARTERS.

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PREFACE.

THIS account of the Shetland Pony is intended to give, in brief outline, the chief facts of the history, and some idea of the present character, position, and use of the breed. The time seems opportune for placing on record some recent phases of its development which might otherwise be forgotten.

We are much indebted for information and help to the Ladies Estella and Dorothea Hope, to Mrs Wentworth Hope Johnstone, to Mr Robert Alexander, R.S.A., to Mr Robert Brydon, to Mr R. W. R. Mackenzie, and to Mr W. Mungall.

The illustration of Norwegian rock-drawings is reproduced from Du Chaillu's 'The Viking Age,' by kind permission of Messrs John Murray & Son.

We are especially grateful that we are allowed to include, as an Appendix to the volume, Mr Cossar Ewart's very interesting discussion of "The Making of the Shetland Pony."

C. D.
A. D.

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The Shetland Pony.

CHAPTER I.

The Early History.

A BREED of small horses appears to have been the first Scottish domestic animal to attract that attention which British livestock now commands so generally. Dion Cassius, as translated by Holinshed, says of the “Calidons,” in the second century of our era, that “they fight in wagons, and have little light and swift horses, which are also very swiftie, and stand at their feet with like stedfastness;”^[1] and “St Austin” is said by Hamilton Smith to describe how “Mannii or poneys brought from Britain were chiefly in use among strolling performers, to exhibit in feats of their craft.”^[2] This race of small horses survives in the Shetland pony.

It has long been regarded as practically certain that the Shetland Islands possessed a native pony before the Scandinavian invasion and settlement of the ninth and subsequent centuries. Hitherto this view has been supported only by the fact that the Bressay Stone—an accredited relic of the period of Celtic Christianity in Shetland—displays a representation of a pony or small horse. Now, however, we are able to rely on a much more definite and conclusive piece of evidence, bones having been found, in the summer of 1911, buried in the kitchen midden of the Pictish *broch* or village at Sumburgh, which are identified (by Professor Cossar Ewart) as part of the skeleton of a pony not more than twelve hands high, and as being of ancient date. This fresh evidence places beyond dispute the fact that the pony was a native of Shetland in very early times.

We also know from rock-drawings, which are so ancient that their origin is lost in antiquity, that horses or ponies were found in Norway at a time lying beyond the beginning of history; and coming nearer to our time, we have clear and definite records showing that in the sixteenth century breeds of small ponies were regarded as belonging characteristically to Norway

and Sweden. Olaus Magnus records that: “There are many Herds of small Horses but they are very strong; for by their strength and agility they exceed many greater bodied Horses; and Forraign and Domestic Chapmen buy them for their pleasure, and transport them into remote lands, to be sold as Wonders of Nature. For they are most ingenious, that they can be taught by them to dance and jump at the sound of the Drum or Trumpet; and it is their Exercise by such shews to get gain. Moreover, they are taught to leap through hoops of Iron or Lead, not very large, as Dogs do, and they will turn themselves about with wonderful swiftness.

“Also being called by their proper names, they do it, more or less, as they are commanded.

“These horses feed, when there is necessity, with nothing but broiled Fish and Fir-tree wood; and they will drink ale and Wine till they be drunk.”^[3]

And again: “The Norway horses are small of stature but wonderful strong and swift to pass over mountains and stony ways; but those of *Sweden* and *Gothland* will travel incessantly, and very swiftly with more meat, over Lakes and high Hills and deep Thickets. But those of Oeland, because they are small, are more for sight than service, though amongst them there are found of a different kind that are notable for labour.

“Also the Finland horses are of good qualities.”^[4]



A SCANDINAVIAN ROCK-DRAWING.

The Scandinavian horses were not all alike in merit, for Gervase Markham says:

“Next, then, I place the Sweathland horse who is a horse of little stature, lesser good shape, but least vertue; they are for the most part pied, with white legges and wall eyes; they want strength for the warres, and courage for journeying; so that I conclude they are better to look upon than imploy.”^[5]

These records, combined with the strong family resemblance between Norwegian ponies and certain types of Shetland pony, lead us to conjecture that there is either some extent of common ancestry in those two breeds or

some cross, near or remote, of one with the other. It is probable that the Scandinavian invaders, whose literature and mythology^[6] as well as their place-names display a deep interest in horses, may have brought horses with them to Orkney and Shetland. With this common element, however, we also find a real difference.

While some Shetland ponies of the present-day closely resemble the Norwegian, there are others which belong to a wholly different type—ponies whose characteristics can only be described by the general term of “Oriental,” long-shouldered, fine-boned, small in head, and with an unmistakable Arab outlook. Such a type as this does not occur in the Scandinavian breeds; and its existence proves clearly the presence in the Shetland pony of some ancestral element not found in the Scandinavian horse. This is all the more clearly shown by the fact that the Shetland ponies of this Oriental type do not form pure continuous or separate strains within the breed, but crop out here and there, sometimes the parents, and sometimes the progeny, of ponies apparently purely Scandinavian. They are evidently reversions to an ancestral type which has deeply influenced the breed as a whole and remains an ineradicable element in it. No facts are yet available to show whether these Shetland ponies of Oriental character could be so interbred as to produce a race breeding true to this type. The attempt has never been made; since the general tendency of recent breeders has been rather to neglect and eliminate this kind of pony.

The existence of this strain in the Shetland pony is undeniable, however we may account for it; but, in attempting to explain it, we are almost entirely in the realm of conjecture. Two possible sources of an actual Oriental cross offer themselves for consideration.

In the year 1150 Jarl Rögnvald of Orkney and Shetland, while visiting Norway, became imbued with the idea of leading a Crusade to the Holy Land; and two years later he set out from Orkney for Jerusalem, arrived there after many adventures, returned by way of Constantinople to Apulia, and travelled thence on horseback to Denmark.^[7]

The Orkneyinga Saga records the journey: “From there they sailed west to Púll (Apulia). Earl Rögnvald, Erling, Bishop William, and most others of their noblest men left their ships there, procured horses and rode to Rómaborg (Rome), and then from Róm until they came to Denmark. From

there they went to Norway where the people were glad to see them. This journey became very famous, and all those who had made it were considered greater men than before.”^[8]

It remains a matter of wholly uninformed conjecture whether these war-worn travellers were so bound in affection and admiration to the equine companions of their journeys and adventures that, instead of leaving them in Denmark, they brought them home to Orkney and Shetland, just as in our day British soldiers brought back to our shores the Basuto ponies that had won their hearts on the African veldt. It is a question to which there is no answer.

We come scarcely nearer to anything that can be accounted as proof when we bring the Shetland pony within the orbit of the vivid and entrancing drama of the Spanish Armada.

Legend has always borne that the Armada, steering its stricken course round the North of Scotland and through the Irish Sea, left horses scattered along the coasts in Shetland, Lewis, Mull, Galloway, and on the Irish shores. The records taken at close quarters come tantalisingly near to evidence; but they never quite reach that level, so far as Shetland is concerned.

It is beyond doubt that a Spanish ship, the *Gran Grifon*, *Capitana* (flagship), was wrecked on the Fair Isle, and that this was the flagship of the Armada de Urcas, commanded by Gomez de Medina.^[9] But “John de la Conido of Lekit in Biskey marriner,” under examination, “saith after the English Fleet parted from them, the Spanish Fleet cast out all the horses and mules into the sea to save their water, which were carried in certain hulks provided for the purpose.”^[10] These hulks (urcas), therefore, contained the horses of the Armada; and the fact that their flagship was wrecked on the Fair Isle seems to bring the Spanish horses to the very coast of Shetland.

Whether they landed on that coast or not we may guess almost as we please. But if they did attain it, what kind of horses were they? The Spanish war-horse of that time, as we find it in the pictures of Velasquez, is much more Belgian than Arab, and by no means a likely source of Oriental type or of any good pony strain. On the other hand, there is considerable weight of legendary evidence in support of the view that horses carried by the Armada made an improvement in British breeds. “The fame of

Newmarket,” says Sheardown, “begins soon after the destruction of the Spanish Armada. Some horses which had escaped from the wrecked vessels are said to have been exhibited at that place and to have astonished those who beheld their extraordinary swiftness.”^[11]

This record suggests that Spanish horses were the source of a distinct improvement in the races with which they were crossed, and especially in the matter of speed; and it is hardly possible to think that this should have been the case unless they were themselves of Eastern breeding.

Apart from these possible sources of an actual Oriental cross in the Shetland pony, there remains the possibility that the original pony of Celtic Shetland was itself similar in type and origin to the Oriental horse, and was, in fact, derived from the same stock which, in other conditions, has given rise to the Arab and the thoroughbred. The investigations of Professor Cossar Ewart^[12] and Mr Ridgeway^[13] point to the strong probability of a triple origin of the horse as it is known to history; and the fact that the Shetland pony, as we have it to-day, is sometimes of a purely Scandinavian type, sometimes of an Oriental type, may perhaps be explained by regarding it as a composite of two distinct races, one having a common origin with the Oriental horse, and the other being identical with the Scandinavian pony. Force is lent to this explanation by the fact that the pony depicted on the pre-Scandinavian Bressay Stone is wholly different in type and character from those represented in Scandinavian rock-drawings, and much more resembles the Oriental horse, with a high carriage and fine type of head, and a short back.

Whatever its earlier history may be, the Shetland pony begins to emerge in definite records during the sixteenth century. Ubaldini wrote in 1568 —“Their horses are very small and tiny in stature, not bigger than asses, nevertheless they are very strong in endurance.”^[14] In the same year Jo. Ben. (John the Benedictine?) speaks of “*alia Insula inculta nomine Auskerrie* [presumably either the Orkney Island of Auskerrie or the Shetland Island Osse Skerry] *ubi equi ferocissimi sunt.*”^[15] These “very wild” horses of the Auskerrie are without doubt progenitors of the Shetland pony of to-day.



THE HORSE ON THE BRESSAY STONE.

In 1576 we find the use of horses by the laird matter of dispute in Shetland. “The Parochinaris of Wais ... deponis that quhen the Laird come throw their parochin, giff the worst boy that was in his companie got not ane horse to ride upon, the Laird wold gar thame that refusit pay 40 babeis thairfair of Zetland payment.”^[16] In 1614 it is recorded by Mackaile that “the horses are little in Orkney”;^[17] while at the same period we have an Act to restrain the grazing of “wyld horsis”;^[18] and shortly afterwards, in 1628, an Act “anent ryding and cutting of other men’s horsis taillis.”^[19]

Within a few years after this the Shetland pony is clearly identified; for Captain John Smith says in 1633: “Their Horses, which they called Shelties, some of which I have seen, are little bigger than Asses, but very durable.”^[20]

From this date onwards we have a continuous record of the pony, growing in definiteness as time goes on. “The horses,” says the Rev. Hugh

Leigh in 1650, “are of a little size and excellent mettell: for one of them will easily carry a man or woman 20 miles a day; and they will live till they be 20 or 30 years of age though they be never stabled summer or winter.”^[21] Travellers comment on its small size, its strength, and its excellence. Thomas Kirke, in his diary, reports a visit to “Burra’s” house (Stewart of Burray in Orkney). “We dined before we went away, having been very well treated, and at our departure he bestowed a little Shetland horse upon us, so low that I could easily stand on the ground with the horse under me.”^[22] The Orkney horses in 1693 are, according to Wallace, “little yet strong and well mettald, most of which they get from Zetland, and are called Shelties.”^[23]

In 1701 we have a full and clear description by Brand which places beyond doubt the fact that the pony whose history we have traced from the vague suggestions of earlier times is the Shetland pony as we have it now.

“I think the kine and sheep are of a greater size than they are in Orkney, though their horses be of a less; they have a sort of little horses called shelties, than which no other are to be had if not brought hither from other places; they are of a less size than the Orkney horses, for some will be but nine, others ten nives or handbreadths high, and they will be thought big horses there if eleven; and although so small yet they are full of vigour and life, and some not so high as others often prove to be the strongest, yea there are some whom an able man can lift in his arms, yet will they carry him and a woman behind him eight miles forward and as many back; summer or winter they never come into a house, but run upon the mountains in some places in flocks, and if at any time in winter the storm be so great that they are straitened for food, they will come down from the hills, when the ebb is in the sea, and eat the sea-ware (as likewise do the sheep), which winter storm and scarcity of fodder puts them out of case, and bringeth them so low, that they recover not their strength till about St John’s mass-day, the 24th of June, when they are at their best; they will live to a considerable age, as twenty-six, twenty-eight or thirty years, and they will be good riding in twenty-four, especially they will be the more vigorous and live the longer, if they be four years old before they be put to work.

“Those of a black colour are judged to be the most durable, and the pied often prove not so good; they have been more numerous than they are now; the best of them are to be had in Souston and Eston, also they are good in Waes and Yell, these of the least size are in the Northern isles of Yell and Unst.

“The coldness of the air, the barrenness of the mountains on which they feed, and their hard usage may occasion them to keep so little, for if bigger horses be brought into the country, their kind within a little time will degenerate; and, indeed, in the present case we may see the wisdom of Providence, for their way being deep and mossy in many places, these lighter horses come through, when the greater and heavier would sink down; and they leap over ditches very nimbly, yea up and down rugged mosses, braes or hillocks with heavy riders upon them, which I could not look upon them, but with admiration, yea I have seen them climb up braes upon their knees, when otherwise they could not get the height overcome, so that our horses would be but little if at all serviceable there.”^[24]

This statement is repeated by Martin in its essential features a few years later. “This Country produces little Horses, commonly call’d Shelties, and they are very sprightly, tho’ the least of their kind to be seen any where; they are lower in stature than those of *Orkney*, and it is common for a Man of ordinary Strength to lift a *Sheltie* from the ground: yet this little creature is able to carry double. The black are esteemed to be the most hardy, but the pyed ones seldom prove so good: they live many times till Thirty Years of Age and are fit for service all the while. These Horses are never brought into a House, but exposed to the Rigour of the Season all the year round; and when they have no grass, feed upon sea-ware which is only to be had at the Tide of Ebb.”^[25]

Brand’s account, confirmed by Martin, completes the series of statements by which we are compelled to recognise that the Shetland pony of to-day is the lineal descendant, with or without some degree of cross-breeding, of a pony which has lived in Shetland from very early times.

The characteristic which most definitely asserts itself throughout all the descriptions, and which is displayed by the Sumburgh bones, is small size; and the significance of this characteristic is greatly increased by the fact

that it remains unaffected by great changes in the conditions under which the pony is reared.

The common and obvious suggestion is that the ponies of Shetland were individually made small by the severity of the conditions under which they lived—that they were and are dwarfs stunted by starvation. But this suggestion is inconsistent with the undeniable result of experience, that the Shetland pony remains small, and indeed shows no tendency whatever to increase in size, when it is reared in Southern climates and generously nourished.

Twenty years ago even so experienced a breeder as Mr Robert Brydon wrote of the South-country studs: “I cannot help pointing out the difficulty their owners will have to contend with in keeping the size within Stud-book requirements.”^[26] Experience, however, has shown this to be a wholly groundless fear. The apparent tendency of the breed in England and Scotland is not to increase but rather to diminish in size: the mainland—bred ponies are not larger but smaller than those on the Islands; and perhaps the present danger is that they may become too small for use and perfect symmetry.

The fact is that there have always been small horses in Britain—at all events in Northern Britain. The remains recently found in the Roman camp at Newstead include horse bones which indicate that the native horses there were from 11 to 13 hands in height. In Shetland there have probably never been large horses.

The size of other horses, originally larger, has been gradually increased, partly by crossing and partly by a deliberate artificial selection, until a sustained effort, forming part of a general agricultural development, has eventually produced the Clydesdale and the Shire horse of to-day. Increase of size has always, of course, been subject to the limits imposed by the available food-supply, so that while the Clydesdale has been of comparatively old standing in the Lowlands, a much smaller horse held the field until quite recently in the Highlands and in Orkney; while, within the Highland area itself, the so-called “garron,” of Perthshire and the richer parts of Inverness-shire, has for its Island counterpart the smaller, harder, and more active Hebridean pony. But it is impossible to explain these variations of size and type as the direct product of liberal or scanty feeding,

although it is no less impossible to disregard the limiting influence of local conditions which prescribe to each district at each period of its development the size and type of horse which can be maintained in vigour within it. Similarly the Shetland pony is not a horse reduced in size by the scarcity of herbage in Shetland. It is the horse whose type and qualities procured its survival in those Shetland conditions which prohibited any considerable increase in its size.

These same conditions fixed other characteristics as well. They prescribed and produced a degree of vigour and robustness fitted for the maintenance of life in adversity, and for the performance of feats of labour and endurance apparently impossible for so small a physical frame: the “mettall past belief” is the mark of a survivor in hard circumstances. They gave a great advantage to individuals sheltered by abundant mane and tail, and, above all, by that waterproof double coat of thick fur and long hair which alone can maintain warmth in wind and rain and mist. They favoured that docility and sweetness of temper which make the Shetland pony more truly domestic than any other horse, because they made it essential that the pony should live in intimate dependence on its owner; and these qualities of disposition find their expression in the small ear and the large soft full eye which are so characteristic of the breed.

The Shetland pony as every one knows it—small, robust, gay, shaggy, alert, strong of bone, short-eared, large-eyed—is the product of natural conditions and human needs in Shetland; and it is a definite race, established by long selection, having characteristics indelibly fixed. It has already been said that within this unity of race there remains real and very considerable variety of type—a variety hardly less great than that which we find between larger breeds of horses; and the fact that the various types do not breed true, but are interchangeable, points to a far-back mixture of races. Yet, in its widely varying developments, the pony remains a fixed breed; and so long as its racial purity is retained its virtues are ineradicable.

CHAPTER II.

The Pony in Shetland.

THE Shetland pony is almost more conspicuous in the simple farming economy of his own Islands than other horses are in British agriculture. He has been the constant theme of travellers and dwellers in Shetland; and their references show that he has been a dominant interest there throughout the whole known history of his home. The statements which have already been quoted are continued and corroborated up to our own day; and everywhere we find the same description of the ponies—their small size and their courage and endurance.^[27]

Typical of many accounts of them is that given by Campbell in 1750. “There are little horses in this Island, which the Inhabitants call Shelties; they are so very small that one may lay his leg over them from the ground; but notwithstanding their Smallness they are both strong and active, and live many years, even till they are blind with Age: I have heard say, some of them live till they are upwards of thirty, and they are never kept within Doors, but are foaled in the Fields, live in the Fields, and die in the Fields. They do little Work, unless it be to carry some Sea Weed, to dung the Ground in the Seed-Time. There is no Horse-hire.”^[28]

Throughout the narratives of eye-witnesses we find everywhere the fact that the ponies are reared and kept in conditions of great hardship. A Highland Society’s report in 1801 tells us that “the horses live in the open fields, summer and winter, night and day, and never get a mouthful except what they can gather, not even when the ground is covered with snow. At the season of labour they are, of consequence, miserable, lean, and weak; so late as the middle of June they are little else than skin and bone, covered with long hair like goats, yet, even in that situation, their spirit is astonishingly great.”^[29]

“They would be more numerous,” says Gifford, “if in any way cared for; but they lie out in the open fields summer and winter, and get no food but

what they can find for themselves; so in bad winters many of them die with hunger and cold. It will, no doubt, be wondered at by strangers that so little care is taken about these sheep and horses which are so useful and beneficial; the reason whereof is, that the poor inhabitants, having used their utmost endeavours, can scarce find food and shelter for their oxen and cows, without which they could not live; and in hard winters many of them die for want of fodder, so they have none to bestow on their sheep and horses, until they find more time to improve the land.”^[30]



BY THE VOE.

It should perhaps be kept in mind, as a qualification of these comments, that while dependent on some degree of help in finding food in winter, and especially so in the poorer parts of Shetland, the pony is much less in need of shelter than most other animals, and appears, indeed, to thrive much better even when he is exposed to severe weather conditions than when he is kept indoors. But the lot of the Island pony is still a hard one in the long winters, when the scanty livelihood which he can gather on the mosses, by

the dyke-sides, and on the sea-shore is but poorly supplemented by the occasional sheaf of oats which is all that his owner can usually allow him.

The great majority of the ponies in the Islands are in the hands of crofters, either owned by them or held on the system of “halvers,” under which merchants or others supply brood mares in return for a half interest in their progeny.

The mares nursing foals are kept usually about the croft until their foals are old enough to follow them to the “Scathold”; other ponies spend their whole summer in the hills, returning to the nearer fields, after the crops are cleared, when the approach of winter makes some additional food necessary.

It is still too common, though less so than formerly, to leave the foals unweaned, with the result that the mares so treated usually foal but once in two years. This wasteful plan is due to the difficulty of finding food for the weaned foals; but the attempted economy so completely defeats its own object that it cannot fail to die out.

In the beginning of last century we find the Highland Society’s report, already quoted, referring to “an absurd custom among the farmers of preserving for stallions ... the most unpromising of the young of the species”;^[31] and very competent observers state in 1845 that “the ponies are now much smaller in size than they were thirty years ago, entirely owing to the fact that all the best and stoutest are exported, and stallions of the most puny size are allowed to go at large.”^[32]

It would appear from this that the selective process by which the small size of the pony has been fixed and exaggerated was not, at this period, one deliberately and consciously promoted, but was contrary to the wishes of those who regarded the interest of the breed, and was the result of economic pressure which encouraged the export of the larger and more valuable ponies, leaving the smaller and cheaper stallions to be employed as stud animals. Larger and not smaller ponies were in point of fact desired; and the decline in size, which seems to have taken place at this period, was a consequence of the poverty and perhaps also of the short-sighted thrift of the crofting owners.

This fact, indeed, sets aside the argument of Mr Vero Shaw^[33] and others, that the breed must always have been kept pure, because no cross could be used to improve it by reducing its size. The temptation to introduce alien blood came from the opposite motive—a desire to increase size; and when we read that in 1788 “a fine young horse of the Norway breed had perished in a marsh,”^[34] we see that the materials for cross-breeding, as well as the motive to practise it, were actually in existence—the results probably remaining in the larger ponies now used for draught in Shetland.

It must be observed that a scarcity of really good stallions, probably arising from the same causes as formerly, is still the chief impediment to the improvement of the Shetland pony in his native home. But this cause no longer operates to reduce size, as fashion has created a demand for excessively small ponies, which tempts the poorer owners rather to sell than to keep them.

In 1865 we have the first record of an actual attempt to reduce the size of the pony, in the very interesting notes on Shetland pony breeding made by “The Druid” in *‘Field and Fern.’*



GOING SOUTH.

“Colonel Balfour, grandfather to the present proprietor of Shapinsay, began pony breeding at the end of the last century. He improved the form; and when the colours did not come as they expected, the natives, with a few drops of whiskey to quicken them, laid the entire blame on Spunky, the Orcadian water-kelpie.

“He was black, they say, and the sire of some of the finest original ponies of the islands; and if he was disturbed in his courtships, he vanished under the waves in a mass of blue flame.

“The Hellersay stock have been quite able to dispense with him, as North Unst has furnished them with some of its choicest jewels.

“Brisk, the chestnut, dates very far back, and headed the Balfour stud for wellnigh thirty years, and his brother Swift was in the flesh for nearly forty-six.

“The piebald Cameron cost £24, and although he rather spoilt the colours, he introduced a better shape, a smaller head, and decidedly truer action. Odin, of the same colour, also kept up the form; Thor got them nearly all skewbalds like himself; and Lord Minimus was a grey and sire of grey beauties. They are shifted from island to island as the grass suits, and require the most careful drafting to keep them at nine hands. Mr Balfour has about 40 in all, of which the majority are duns and creams; and they are always broken at three, and made very tractable in a week. Her Majesty has a pair of them; and some of the more fancy colours were once picked up by Ducrow.”^[35]

Colonel Balfour, whose enterprise is referred to by “The Druid,” was probably the first to attempt breed improvement in the Shetland pony. His grandson, in “The Druid’s” day, was in all likelihood the first breeder who made a systematic and deliberate effort to accentuate the small size which the poverty of nature and man had already fixed as a breed-characteristic; and his example has not been very widely followed in Shetland.

It cannot, in fact, be said that, on the whole, any clear idea dominates the plans or purposes of pony breeders in the Islands. Individual breeders here and there have pursued an enlightened course in endeavouring to improve their herds; and it is natural that their choice of breeding stock should have been determined largely by the nature of the commercial demand. They have thus been led to concentrate their attention mainly on the production of animals with the weight of body and strength of bone which have been demanded by British and foreign buyers. On the other hand, the conditions of existence in Shetland have greatly contributed to the preservation of an active type of pony such as can gain its livelihood on the poor and mossy pastures of the Islands.

It must be remembered that in many districts there has been, as has already been said, a great dearth of good sires, so that selection of suitable breeding stock has been difficult, and mating has often been carried on, of necessity, very much at haphazard. It is thus all the more remarkable that the pony, so long neglected and so little cultivated in its home, should display so high a degree of excellence as it does. Much of the credit of this belongs to the Marquis of Londonderry, whose stud in Bressay, under the charge of Mr Brydon and Mr Meiklejohn, developed a strain of ponies which fixed many of the best qualities of the breed and became a potent

centre of its improvement; and no account of the Island ponies would be complete which did not mention the successful activity of Messrs John Anderson & Sons of Hillswick, the late Mr Bruce of Sumburgh, Mr Anderson Manson, and the Messrs Sandison. Notable throughout Shetland, the fine quality of the pony is specially conspicuous in Unst, which still retains the superiority which “The Druid” found in it in 1865. “The best ponies come from Unst; but both there and everywhere the breeders are far too indifferent to the points of a sire, as long as they are foal-getters. About a quarter of Unst has a skeleton of red sandstone and serpentine, with a thin soil studded with large red stones and the knobs of rock sticking up. Yet among these rocky incumbrances one sees scores of ponies picking the green grass, which the light of Heaven and the breath of the Gulf Stream force up from so barren-looking a bed. Still, Unst may be regarded as the heart of Shetland; and a sunny, genial-looking spot it is, when other parts of the country are dismal enough, in the late northern spring. The heather and the bog-grasses elsewhere do not make much milk, and the mare ponies sink so much in condition that they are invariably barren every other year. If well kept they reach 44 inches; but the average is from 38 to 42. Their owners frequently lose sight of them for a couple of summers, and recognise them when wanted, not by any formal ‘Exmoor Brand’ on the saddle place or the hoof, but by a peculiar slit or bits of tape, clout, or leather tied through a hole in the ear. Each cottar has generally a few ponies on the hill, and when the May and October sales at the different stations are at hand they circumvent them for a selection by the dealers with a line of forty or fifty fathoms. Still, the hard-working Shetlander is little more than nominal lord of his pony: poverty is his lot from the cradle to the grave, and, as the phrase goes, he is ‘still in tow.’ In his dire need the merchants become his mortgagees, just as the curers are to the herring-fishers: they advance money on the security of his foals, and he doesn’t get the best of it with ‘halvers’ mares.”^[36]

The chief defects of the Island ponies are to be found in the movement and conformation of the hocks—“cow hocks” being common, and also a tendency to excessive bending of the joints. There is, in fact, a look of “curbiness” about many of the ponies which renders it surprising that curb itself—like almost every other unsoundness—occurs but rarely. How far these hock defects are caused or aggravated by undue hardship in early life cannot easily be estimated, but they can certainly be greatly mitigated by

more generous treatment. Apart from them—and from a tendency to roach backs, undoubtedly aggravated by poor rearing—the Island ponies present few common defects that are practically serious; but their general appearance is often much deteriorated by insufficient care in early life.

In colour the pony is much more variegated in the Islands than on the British mainland, where black and brown increasingly predominate. In some parts of Shetland—notably in the western district of Sandness—piebalds and skewbalds are more common than self-coloured ponies; while chestnuts, yellow duns, and mouse-duns (sometimes curiously called “greys”) are exceedingly frequent.

But we still find as “The Druid” did in 1865: “Duns are in great request; but the colour is not so much an object if the bone be only good. Greys and chestnuts are scarce; bay has not its wonted supremacy; and bays and blacks are most common. Some buyers began to go against piebalds from a belief that they had Iceland blood in them, and were softer and slower in consequence.”^[37]

The employment of Shetland ponies in Shetland is now much less than it was formerly. Speaking generally, they have become a breeding stock, kept for sale rather than for work. Somewhat larger ponies—from 11 to 12½ hands—are in very common use in carts; and these are probably cross-bred ponies partly of Shetland ancestry. But the introduction of wheeled vehicles in the latter part of last century almost made an end, in practice, of the pony as a means of transport in its own home.



COMING FROM MARKET.

The fact—apt to be forgotten in controversies about Shetland pony type—is that the pony never until quite recently was a draught animal. Roads did not exist in Shetland until they were made, in and after 1847, in order to give employment for the relief of distress caused by the potato famine. Till then wheeled vehicles were practically unknown, and the ponies were used only as pack and saddle animals. We read of them “travelling through the country among the rocks and mosses”;^[38] and Edmonstone gives us a luminous glimpse at once of Shetland society and of the stature of the riding ponies:—

“Winter is the season of general mirth and festivity in Zetland, although the wish to visit each other is greatly interrupted by the difficulties which are attendant on travelling. As there are no regular roads, a journey over land is a serious undertaking, for the ground is wet and unequal and the ponies are low.”^[39]

One seems to see the cavalcade picking its way through the moss, riders holding up their feet to avoid the soft ground through which their mounts find a path, and ladies tremulous over the fate of the precious burdens of the pack-ponies.

Hibbert gives us an even more complete picture of the Shetland pony in use a hundred years ago:—

“A walk through the valley near Woodwick leads to a large open lawn at the end of the Loch of Cliff, which seemed very populous and well cultivated. I arrived there on the Sabbath morning; the natives of the Vale were all in motion in their way to the Kirk of Baliasta. The peasant had returned home from the bleak scathold, where he had ensnared the unshod pony that was destined to convey him to the parish kirk. No currycomb was applied to the animal’s mane, which, left to nature’s care, ‘ruffled at speed and danc’d in every wind.’ The nag was graced with a modern saddle and bridle, while on his neck was hung a hair-cord, several yards in length, well bundled up, from the extremity of which dangled a wooden short-pointed stake. The Shetlander then mounted his tiny courser, his suspended heels scarcely spurning the ground. But among the goodly company journeying to the kirk, females and boys graced the back of the shelty with much more effect than long-legged adults of the male sex, whose toes were often obliged to be suddenly raised for the purpose of escaping the contact of an accidental boulder that was strewed in the way. A bevy of fair ladies next made their appearance, seated in like manner on the dwarfish steeds of the country, who swept over the plain with admirable fleetness, and witch’d the world with noble horsemanship. The parishioners at length arrived near the kirk, when each rider in succession, whether of high or low degree, looked out for as green a site of ground as could be selected, and, after dismounting, carefully unravelled the tether which had been tied to the neck of the animal. The stake at the end of the cord was then fixed into the ground, and the steed appeared to be as satisfactorily provided for during the divine service as in any less aboriginal district of Britain, where it would be necessary to ride up to an inn, and to commit the care of the horse to some saucy lordling of the stables.”^[40]

Peat-carrying appears to have been one of the main duties of the pony in the early part of last century.

“It appears that the use of the shelty, which is seldom more than from nine to eleven hands high, is principally confined to the carrying home of peat; yet, in the transportation of other kinds of light burdens, his back is still surmounted with a wooden saddle. When hay or any light bulky substance is to be carried, *maiseys* are used, which are made of ropes prepared from *floss* or rushes, these being reticulated in meshes of some inches in width. A net of this kind is passed round the horse, so as to secure the hay or other light substance that rests upon the boards of the *klibbar*. This ancient saddle is also found of use when the shelty is required by the female rider to bear her to the parish kirk; she then throws over his back a native coarse manufacture of the country, woven into the shape of a saddle-cloth, and when upon this covering the *klibbar* is fixed, its projecting pieces of wood which the female holds by, form it into a kind of sidesaddle.”^[41]

Till recent times, long after the ridden shelties had given place to the road-using gig, ponies were almost universally employed as carriers of peat.

Cowie writes in 1874: “The peats are now dry, and are either built into a stack on the hill, thence to be gradually removed in *cassies* during the year, or are immediately conveyed home on the backs of ponies, or in carts. The apparatus by which the pony is now literally turned into a beast of burden consists of a pair of straw panniers or *maysies* attached to a wooden *clibber*.

“This process of transport is termed leading the peats. Long strings of ponies engaged in this way may be seen in the month of July, under the command of *peat boys*.”^[42]

In the remoter Islands, ponies are still to be seen carrying creels of peats: but even this is now an extinct use in most districts. The pure-bred pony in the Islands has never been a draught animal to any great extent; and with the introduction of wheeled conveyances its employment has almost entirely disappeared.

Another ancient use—in a sense a by-product—of the pony has also ceased. We find in the old laws of Shetland not merely prohibition to “ride, labour, or use any other man’s horse without liberty of the owner,” but also to “cut any other man’s horse—tail or main—under the pain of ten pounds.”^[43]

Thus did the horse-owning fisherman protect the material of his lines. But this use of the pony became extinct even before the changes had set in which are relegating line-fishing to the region of dead industries.



CARRYING PEATS.

Yet, although the local employment of the pony is a thing of the past, its production remains a profitable part of Shetland farming.

The early records show very low prices for ponies. The ‘Statistical Account,’ 1845, places them at from £1, 10s. to £5; “The Druid,” in 1865, sets the value of horses at £7, and of mares at £5;^[44] and even in 1871, when the coal-pits had been using Shetland ponies for twenty years, Cowie valued the horse at £8 to £10, and the mare at £3 to £5. Such prices bear no relation to those of recent years, when good stallions have realised from £18 to £20 for pit work, while better ponies, when fully pedigreed, command very much larger prices. At the earlier rates ponies can have yielded but a poor profit; but under recent conditions they must give a large return on the

comparatively small cost of breeding and rearing them in Shetland; and this increasing profit encourages the hope that crofters and others in Shetland may be more energetic in the future than they have been in the past, in improving the pony which is one of the best assets of their Islands.

THE FETLAR PONY.

The Island of Fetlar contains, besides pure Shetland ponies, a distinctive breed of its own.

The Fetlar pony is not, indeed, of pure race. Its origin is traced to an animal which has often been called a "mustang," but was in point of fact a grey Arab, presented by the famous General Bolivar to the late Sir Arthur Nicolson. From 1837 onwards, for some years, this horse was crossed with the native ponies of the Island, which were presumably somewhat larger than those of other parts of Shetland, Fetlar being one of the best grazings in the Islands. For many years the influence of this horse showed itself in a large proportion of grey ponies in Fetlar. Later, another Arab was introduced; and an Orkney garron cross was also used. The resulting product is a pony of about twelve hands—ranging from eleven to thirteen—from which the grey colouring is now practically eliminated. The ponies bred in this way show a degree of unity in type and size which is truly remarkable, in view of the apparently very different sources from which they are sprung. Indeed, the singular fidelity to type of this essentially cross-bred pony is highly suggestive of a common element in the ancestry of the Arab and the Shetland pony. The Fetlar is, in type and quality, one of the best of native ponies. It has a good deal in common with the ponies of the outer Hebrides.

CHAPTER III.

The Modern Pony.

THE Shetland pony, as now produced on the British mainland, is chiefly derived from the stud established by the Marquis of Londonderry in the Islands of Bressay and Noss in 1870,^[45] and carried on by him there under the management of Mr Robert Brydon and the late Mr Meiklejohn until its dispersal in 1899. It was in this stud that the standard was set by which showyard judging has proceeded during the last twenty years; and it was here also that the type of the modern pony was created and fixed by selection and close inbreeding.

The stud was originally intended for the purpose of breeding pit-ponies for its owner's collieries in Northumberland; and this purpose is reflected in the type which is characteristic of the "Londonderry" pony. It is commonly said that the governing formula of the stud was "as much weight as possible, and as near the ground as it can be got"; and, so far as it goes, this formula—admirably adapted as it is for defining a pit-pony—is no bad description of the result attained in many instances. While the original object of the stud was never lost sight of in its policy and management, it was carried out by skilful and enthusiastic breeders, who set themselves to eliminate defects of conformation which were common among the Island ponies. The consequence was a degree of breed improvement which is perhaps without a parallel as the result of less than thirty years of breeding and management.

But not only is the source of improvement in the modern pony to be found chiefly in this one stud. It is also traceable—as in so many other breeds—mainly to a single animal.

If it be true that the modern pony is substantially the "Londonderry" pony, it is hardly less true that the "Londonderry" pony is the pony that is bred from the horse Jack (16), foaled in 1871, which came into Lord Londonderry's possession as a colt, and was the sire or grandsire of almost

all the stallions used in the stud, as well as of a third of all the mares that are recorded as belonging to it.

Some idea of the extraordinary predominance of Jack in the stud may be gathered from the entries in the first nine volumes of the 'Shetland Pony Stud-Book,' in whose tenth volume the dispersal sale of the stud at Seaham Harbour is recorded. We find that in the period covered by these entries, nineteen stud horses were used to a greater or less extent. Of these, in addition to Jack himself, there were his three sons, Laird of Noss (20), Lord of the Isles (26), and Odin (32); and his eight grandsons, Thor (83), Sigurd (103), Emeer (131), Runolf (62), Najal (75), Lava (121), and Otkell; while Oman (33), his great-grandson on the dam's side, was also considerably used. We find also that these ponies related to Jack were much more extensively bred from than the unrelated sires, so that of the four hundred and ninety foals entered as produce, two hundred and forty-eight are by Jack and his three sons, and a hundred and sixty by his eight grandsons; while thirty-six are by Oman, and only forty-six by sires wholly unrelated to Jack.



JACK (16).

An analysis of the list of dams bred from in the stud accentuates the meaning of these figures; since out of a total of a hundred and twenty-five entered in the Stud-Book, seventy-six are by Jack and his three sons, while ten are by his grandsons.

The result of this selection of breeding stock appears in the extent to which the progeny are inbred to Jack. Of the hundred and twenty-five mares, fifty are sired by Jack, his sons, and grandsons, out of dams similarly sired; while forty are sired by Jack and his three sons, out of mares sired by them. Of the four hundred and ninety foals entered, two hundred and eighty-two are sired by Jack and his sons and grandsons, out of mares by the same list of sires.

Of the fifty-one mares not bred from Jack and his direct descendants, twelve were daughters of Prince of Thule (36), and thirty-six by his son Oman.

Prince of Thule is thus, next to Jack and his sons, the most important sire element in the stud. His influence, however, is considerably reduced by the almost exclusive extent to which he was mated with mares sired by Jack and his sons. Of the twelve mares sired by him, seven are from daughters of Jack, one from a daughter of Odin, and one from a daughter of Laird of Noss; while twenty-four of the dams of his thirty-four foals are daughters of Jack and his three sons.

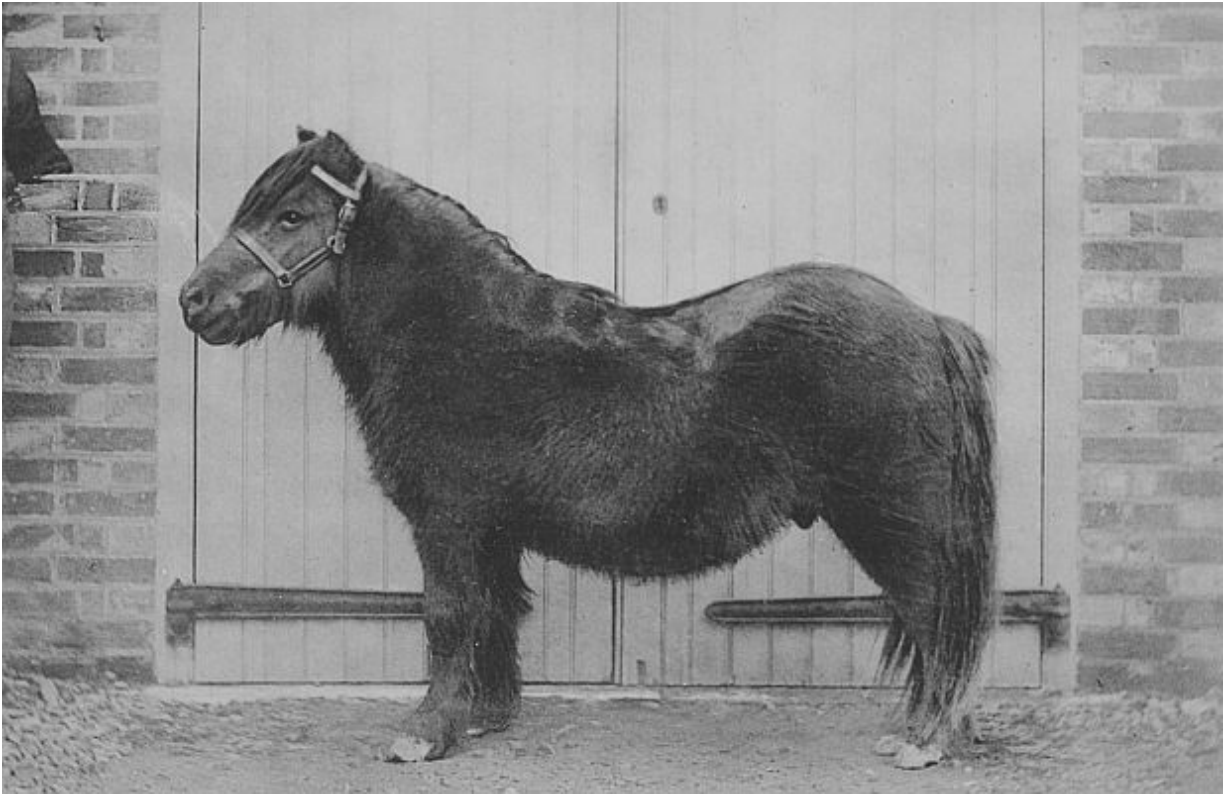
The influence of his son Oman is similarly discounted, five of his six daughters bred from being daughters of Jack's sons, while twenty-eight of all his thirty-six foals have dams similarly bred.

It is probable that the results of an out-cross of this kind upon an inbred stock like that created by the continuous use of Jack and his sons will be slight and transient; and in point of fact the general influence of Prince of Thule upon the Londonderry Stud has not been greater than might have been expected.

The singular predominance of the Jack race in the modern pony is illustrated by the showyard results of recent years.

At twelve shows of the Highland and Agricultural Society held since the Londonderry Stud was broken up in 1899, a hundred and sixteen first and second prizes have been awarded in Shetland pony classes. Of these, a hundred and fourteen have been gained by the progeny of sires actually in use in the Londonderry Stud: Laird of Noss, Lord of the Isles, Odin, Oman, and Thor; and of their sons and grandsons. Laird of Noss, his sons Harold (117), Duncan (147), and Hector (183), and his grandson Merry Hero (244), sired fifteen; Lord of the Isles, his sons Multum in Parvo (28), Sigurd (103), Vespa (166), and Naughty (204), and his grandson Rattler (210), twenty-one; Odin, his sons Olaf (59), Bonaparte (168), Uniacke (177), Palmer (228), Besieger (235), Diamond (257), and Peace (325), and his grandsons Monkshood (274) and Norman (276), twenty-eight; Oman, his sons Frederick (223) and Seaweed (333), and his grandson Glencairn (314), twenty-five; while Thor was himself sire of twenty-five; and it is noteworthy that the winner of the two distinctions not gained by the offspring of Londonderry sires was bred from a dam inbred to Jack's grandson Multum in Parvo. Taken as a whole, this practically exclusive domination of the showyard, for twelve years, by twenty-six sires of

Londonderry origin, of which three are sons and fifteen are grandsons of Jack, while the remaining eight are descended from him, and all of them without exception are otherwise closely related to his stock, is a remarkable demonstration of the influence of a single stud and a single horse upon the breed as we have it to-day.



ODIN (32.)

Of Jack's parentage nothing is known. It is probable, however, that he was himself an inbred animal; for close inbreeding is still, and always has been, the general practice among pony breeders in Shetland, probably rather through necessity or carelessness than as the result of deliberate intention; and Jack's prepotency as a sire lends colour to this supposition.

He was a black, 40 inches high, and the only portrait of him which we possess shows him to have been a short-backed and close-coupled horse of remarkable bone and substance, finely proportioned, and with a bold and upright carriage. He must have had a sound and vigorous constitution, since he lived to the age of thirty, and was at stud to the end of his life.

His most famous son, Odin, also a black horse, was 38 inches high. Odin's dam, Nugget, was sired by Tom Thumb (44), whose height is stated as 34 inches, and who was brought back from work in the pits in 1879 with the view of producing ponies of small size.

Odin was a horse of immense power and robustness, and great masculinity of appearance. His bone and weight were his most salient characteristics; but he was a vigorous and active mover, with strong hock action, though not perfectly straight in his going. He was disfigured by a head heavy even out of proportion to his general bulk and weight. He was probably the most successful sire among Jack's sons, his male descendants being conspicuously better than the females.

Laird of Noss, also a son of Jack, was a black pony, 38 inches high, with some white marks on his near side. He was a pony of somewhat lighter build and more upstanding carriage than Odin, with a finer head and less bone. He is best known as the sire of the famous horse Harold, of Duncan, and of Hector; and it is through them that his strain is perpetuated.

Jack's other son, among the Londonderry sires, was Lord of the Isles, a pony which, like his brothers, was black, but was two inches smaller than they, his height being given as 36 inches. He was a thick and compact pony, less used in the stud than Odin, who appears to have been most approved by the stud management, as only sixty-three foals by Lord of the Isles are recorded against a hundred and nineteen by Odin. Lord of the Isles is chiefly interesting as the first instance of the introduction into the blood of the Londonderry sires of the cross of Prince of Thule, which has already been referred to. His dam Handy was a daughter of Prince of Thule; and it may perhaps have been from this source that his reduced size came, for Prince of Thule was himself only 36 inches high. It is significant, too, that Multum in Parvo, a brown horse 37 inches high, sired by Lord of the Isles out of his own dam Dandy, is described by those who knew Prince of Thule well as being exceedingly like him.



MULTUM IN PARVO (28.)

Multum in Parvo is probably the best-known son of Lord of the Isles, whose blood is otherwise mainly transmitted through the descendants of his daughters. He died in 1912 at the age of twenty-eight, having been foaled in 1884. His crest had latterly fallen over; but he still retained a singular air of distinction and a picturesque quality hardly to be discerned in many of the more massive ponies. He must always have lacked power and weight and strength of action; but his look of breeding, his quality, and the magnificent abundance and straightness of his curtain-like mane and forelock, attested an element in his breeding which should not be lost sight of.

This estimate of Lord of the Isles as a sire is borne out by the conspicuous qualities of his daughter Boadicea (998). This beautiful black mare, 36 inches high, is no doubt somewhat deficient in bone. But she stands almost by herself among Shetland ponies as an example, approaching closely to perfection, of what a riding-pony ought to be, with a small and exquisitely shaped head carried high on a clean-cut and well-

arched neck, shoulders that would not disgrace a good thoroughbred, fine withers and short strong back, and the safe and easy action that properly belongs to an animal of her type.

Lord of the Isles' name appears in the pedigrees of a large proportion of the best Shetland ponies, especially through his famous daughter Beauty (167); but apart from every other claim that he may have, and in spite of his perhaps too limited use in the Londonderry Stud, the fact that he is the sire of Multum in Parvo and of Boadicea entitles him to rank as a stallion of the first importance.

With these sons of Jack used in the Londonderry Stud must be mentioned his grandson Thor, now the sole survivor of the original Londonderry stallions. He is a son of Odin out of Fra (185), and is a brown horse 38 inches high. Like Lord of the Isles, he is related on his dam's side to Prince of Thule, her sire; and he represents, therefore, almost the same combination of strains as Lord of the Isles. At twenty-seven years old he retains in a remarkable degree the vigour and vitality of youth. Slightly grey now over the cheekbones, and fallen a little in his spine, he still holds his crest erect, and moves with freedom, speed, and gaiety. He is perhaps a little larger in head than is desirable; but he is a pony of great substance and power, with abundance of well-shaped bone; and he displays very pure Shetland character. He was freely used in the Londonderry Stud, fifty-six of his foals being entered in the Stud-Book—a larger number than is credited to any other stallion except the sons of Jack.



THOR (83.)

He has hitherto excelled as a sire of mares rather than of stallions—Beatrice (1533), Bracelet (1604), Perfection (1489), and Stella (1496) being among the most famous of his daughters. For about six years after the Londonderry Stud was broken up he appears to have been comparatively little bred from; but during the last six seasons he has been more largely used.

A still greater degree of the Prince of Thule cross was introduced in Oman, a dark-brown horse 34 inches high. Oman was a son of Prince of Thule; and his dam Norna was a daughter of Lord of the Isles, himself, it will be remembered, a son of a Prince of Thule mare. Oman was a compact and massive pony, showing great quality and good action. Among his best-known sons have been Frederick (223) and Seaweed (333); while he was the sire of such mares as Belle of Bressay (1192), Sea Serpent (1535), Silver Queen (1197), and Harriet (1194).

Prince of Thule represents the one considerable element in the Londonderry Stud which was unconnected with Jack, and tended perhaps to counteract his influence. He is described by those who knew him as a pony of exquisite quality, with a small thoroughbred head, prominent wide-set reddish hazel eyes, and an exceedingly fine muzzle. He was short-backed, with strong quarters, somewhat inclined to droop, but finished with a well-carried tail; and he was somewhat cow-hocked. He had big wide feet; and his bone was strong, with large joints. His rein was long and his withers high, though his shoulders were somewhat straight; he was a conspicuously close mover. In colour he was a seal brown, with very bright tan muzzle and flanks; and his mane hung to his knees and his forelock below his nose.

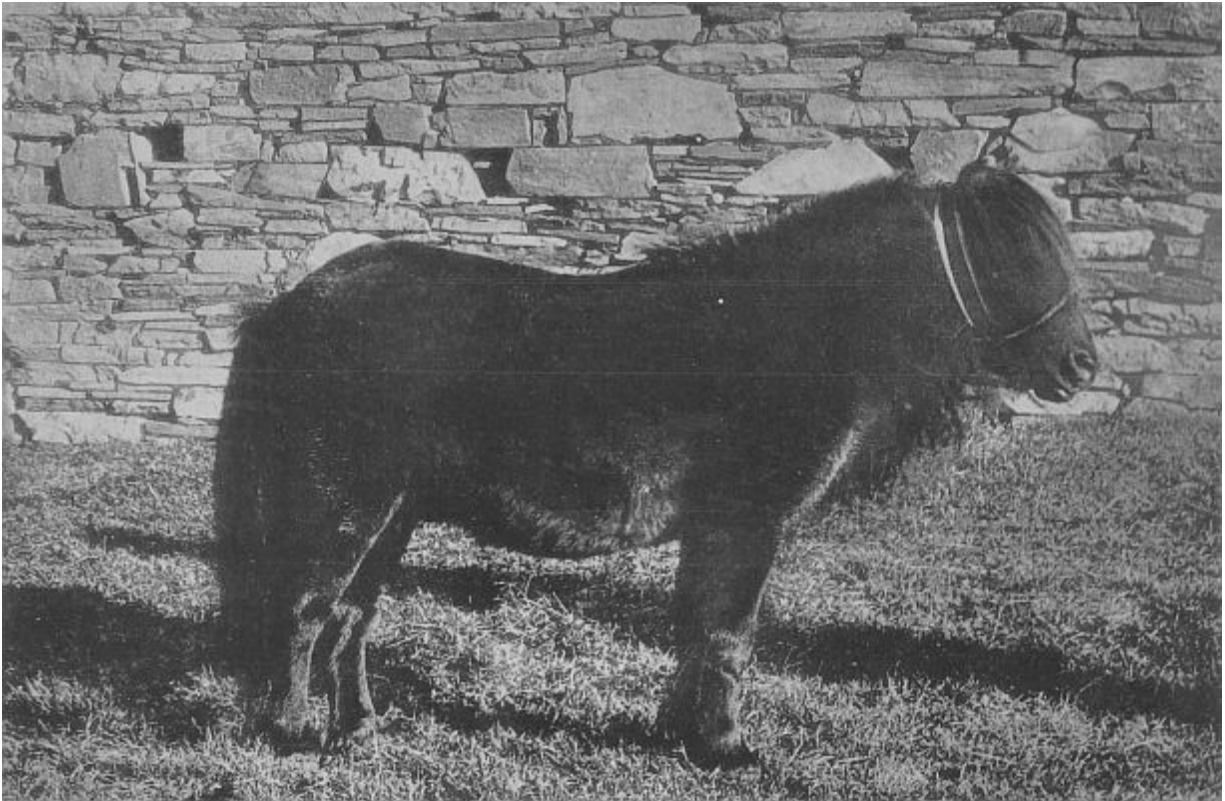
Another sire which had a much less important place in the stud was Lion, a dun pony 36 inches high, bred by Mr Bruce of Sumburgh. He is described as a well-coupled pony, but rather long and low.

It will be recognised that Prince of Thule is the source, and probably the only immediate source among the Londonderry sires, of whatever may be found in the modern pony to represent that "oriental" type which it has already been said has all along been an integral element in the Island pony. The Jack blood is mainly, if not indeed exclusively, that of the Scandinavian type as opposed to the other type, which is depicted on the Bressay stone, and which is represented in modern times by Multum in Parvo and Boadicea. In Prince of Thule, and whatever impression he may have made on the pony of to-day, is to be found the main source, within the Londonderry strain, to which those must turn who desire to produce the riding as distinct from the draught or pit-pony. It has already been said that his influence is largely counteracted by the extent to which he and his son Oman were mated with mares by Jack and his sons. But his stock remain the best hope of breeding ponies which should combine the many excellences of the typical "Londonderry" pony with the quality and activity in which it is apt to be somewhat lacking. It is worthy of note that the combination of Prince of Thule and Odin blood has always produced a large proportion of good foals.

When we come to examine the female lines in the Londonderry Stud, we find, as has already been noted, that they are largely the produce of the sires which we have just reviewed; and we find also that a great proportion of the best animals produced trace from a few of the original mares. No complete

analysis in this respect can be attempted; but it will be found that four mares bulk largely in the formation of the stud.

- I. Darling (174), by Jack (16), was the dam of Darling II. (175), by Laird of Noss, of Dixie (664), by Odin, and of Beauty (167), by Lord of the Isles—Beauty being the dam of Besieger (235) and Bretta (811), both by Odin; while Bretta became the dam of Beatrice and Bracelet by Thor, and of Belle of Bressay (1192) by Oman.
- II. Fra (185), by Prince of Thule, was the dam of Thor by Odin, of Harold by Laird of Noss, of Frederick by Oman, and of Hildigunna (668) by Lord of the Isles.



PRINCE OF THULE (36.)

- III. Peggy, by Jack, was the dam of Pride (202) and Princess (203), both by Prince of Thule—Pride being the dam of Petite (1196) by Odin, and Princess being the dam of Pansy (1282) by Oman, and of Perfection (1489) by Thor.
- IV. Swertha (211), by Odin, was the dam of Silver Queen (1197) by Oman, and of Sigurd (103) and Sweetie (676), both by Lord of the Isles—Sweetie being the dam of Strawberry (1635) and Sapphire (1276), both by Odin, and of Snowdon (1112) by Thor.

These are the leading mares of the stud. It will be observed that two of them are daughters of Jack, one of his son Odin, and one of Prince of Thule;

while two of their dams are daughters of Jack, one of his son Lord of the Isles, and one of Prince of Thule.

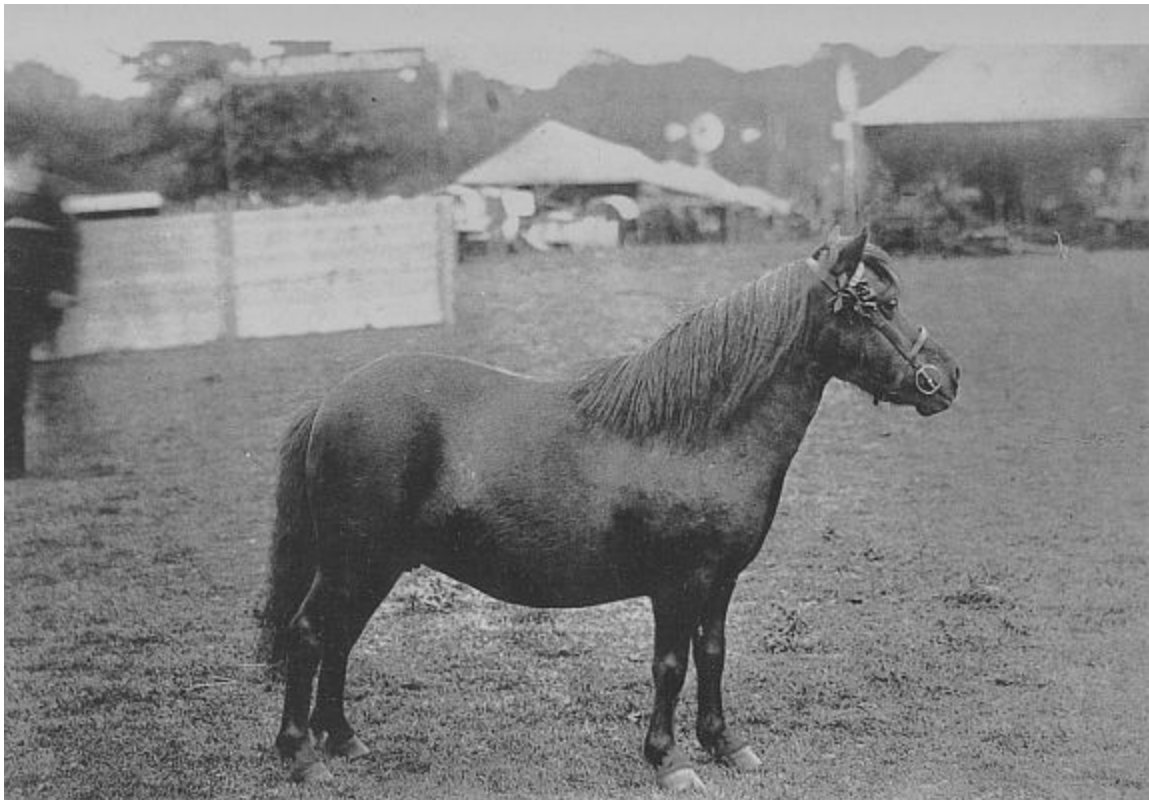
The development of the Shetland pony in the studs of present-day breeders has been greatly advanced by the 'Shetland Pony Stud-Book,' which was first issued in 1890, and which has since been published annually with increasing usefulness and success. The twenty-three volumes which have now appeared form a record invaluable to breeders. Dissatisfaction on the part of certain breeders in Shetland, who found out too late the ill effect of neglecting to enter their ponies (a dissatisfaction stimulated perhaps by interested persons), has recently led to the formation of a '*Shetland Island Pony Stud-Book*,' registering only ponies bred in the Shetland Islands. This book, however, is of very secondary interest and importance, and its restricted scope naturally prevents it from becoming a complete Stud-Book of the breed. Multiplication of Stud-Books is evidently disadvantageous to a breed; and it is to be hoped that the Stud-Book proper may soon become once more the combined record of all pedigreed ponies in Shetland as well as in other parts of Britain.

Useful Stud-Books of the breed are published in the United States of America and in Canada.

The pony as we find it in the showyards of Britain and in the studs of the principal breeders exhibits, in the main, the characteristics of the Londonderry strain.

Its size remains, as it always has been, its most marked characteristic. No ponies over 42 inches in height are admitted to the Stud-Book; and a height of more than 40 inches is properly regarded as a serious fault. This, indeed, is a matter of vital importance, since any considerable increase of size deprives the pony of its individuality and brings it into comparison with other breeds. On the other hand, there has, in recent years, been a tendency to undue diminution of size—the former desire of breeders to increase height having given place to a morbid ambition to produce pigmy ponies. It must be kept in mind that ponies of sizes less than 34 inches are of little use for practical purposes, since, unless they are quite disproportionately massive, they cannot have sufficient weight and strength for draught, while the undue shortening of their legs deprives them of the leverage and activity necessary for saddle or harness. It must also be remembered that the

cannon-bone cannot be shortened beyond a certain point, and that any exaggerated shortness of the fore-leg is therefore only to be obtained by a disproportionate reduction of length of arm, fatal to symmetry and productive of cripples. Anything which tends to make the pony merely an oddity and a toy, and to take it out of the category of useful or usable horses, is fatal to the prospects of the breed and should be resisted by breeders and judges.



SAPPHIRE (1276.)

Speaking generally, about 38 inches is the height which will be found, while retaining the individual character of the breed, to lend itself best to symmetry and activity. There are, no doubt, excellent ponies both larger and smaller; but a study of the recorded measurements of the best animals will uphold the view that this is the height at which, speaking generally, the pony can be produced at its best.

The most salient and essential feature of the Shetland pony, next to its size, is its general air of hardihood and vitality. Stamina and robustness—

capacity to endure both work and hardship—are among its most essential merits; and they should appear in its demeanour, displayed in spirit, boldness, and a high though docile and generous temper. Ponies which show feebleness of appearance or constitution should be rigidly excluded from the stud and the show-ring, and soundness should be made essential.

The Shetland pony is one of the soundest of horses—bone defects being almost unknown in it; and a vigilant watch should be maintained against anything that might impair its character in this respect. But second only in importance to physical soundness is the temper and disposition of the pony; and sluggishness and a lethargic mien should be counted as serious faults.

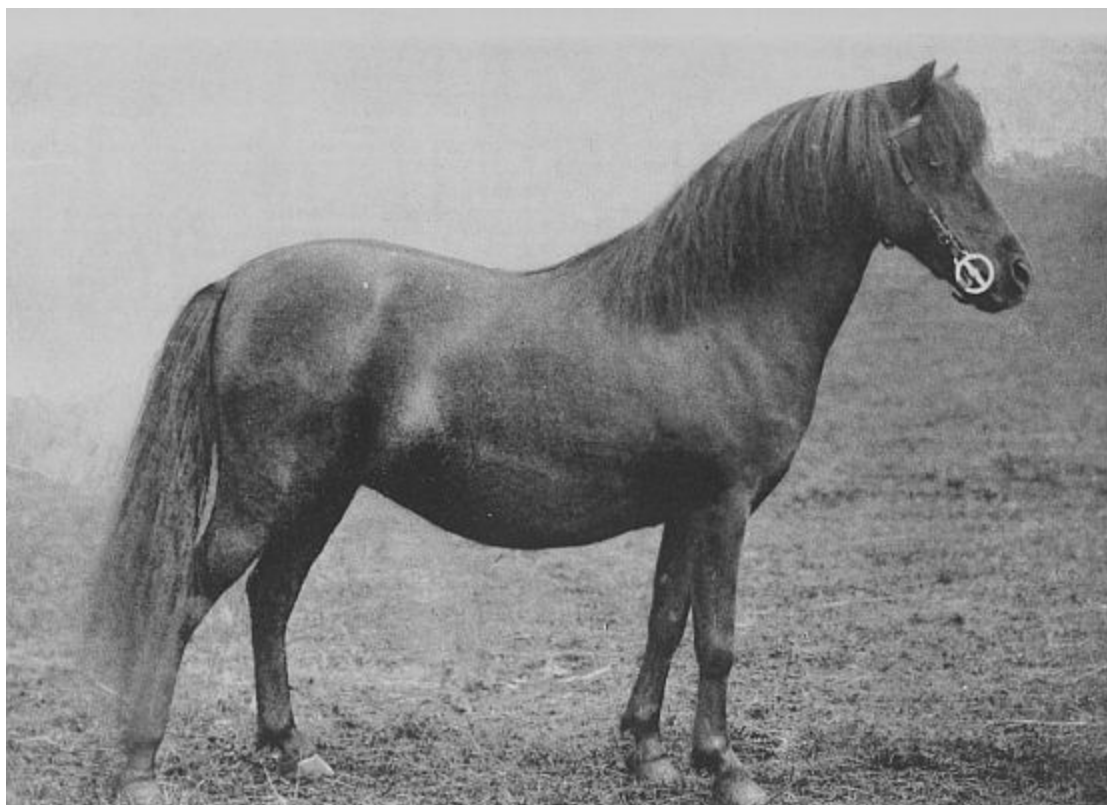
The general symmetry of all good horses is very much the same; and the Shetland pony is no exception. But, in particular, he should be deep through the heart, short and strong in back, well ribbed up, and well sprung in barrel. The shoulders should be long and well sloped, showing sharp and distinct withers. The quarters should be broad and long, and well filled up, with the tail set high and carried gaily. The neck should be long, well arched, and powerful, fine at the gullet, and carrying the head high and well forward.

Perhaps in no point is the present-day Shetland pony so often defective as in shoulder. Many of the most substantial and characteristic ponies of the “Londonderry” strain are short and straight in shoulder and wholly lacking in withers. Such ponies as these may be useful in the coal-pits, but they are useless above ground. They can never be really fine and active movers; and they can never be—what the Shetland pony ought to be—the child’s riding-pony. Shoulders and withers that will hold a saddle should be regarded as a *sine quâ non* of a really good pony. In this respect much yet remains to be done in the improvement of the breed. In other directions great advance has been made; and long backs, flat sides, and short and drooping quarters are less in evidence every year. The general style and symmetry of the pony are steadily improving.

Nothing in the proportions of a pony more affects his appearance than the size, form, and carriage of his head. Undoubtedly many of the most massive and powerful of present-day Shetland ponies are disfigured by heads which are not merely out of proportion to their size, but which are also carried much too low. This last defect commonly arises from; and goes with,

defective shoulders; but, from whatever cause it proceeds, it must be regarded as a most serious blemish, fatal alike to the appearance of the pony and to his safety and pleasantness as a mount. To eliminate it from the breed without sacrificing the substance and power with which it is often associated may be a matter of skill, time, and patience; but breeders ought not to be satisfied until this object has been attained.

The head itself should be small and short, wide across the forehead, relatively long from ear to eye, with a muzzle short and fine and somewhat hollowed, or almost “dished,” immediately below the eyes, which should be large, full, and prominent, looking well forward, so as to be clearly seen from in front. “Ringle” or “wall” eyes are a serious though not a common defect and should be discouraged. The ears should be small and erect, wide set, but pointing well forward, the nostrils wide and open. The shape and carriage of the head are even more important than its size.



BOADICEA (998.)

It is perhaps in limbs and joints that the modern pony marks the largest advance upon his unimproved Island forefathers. Reference has already been made to the apparent “curbiness” of the hocks of many Island ponies. This is a defect that has very largely been bred out of the Londonderry strain, in which good joints, and particularly good hocks, have, with occasional exceptions no doubt, become well established. Strong and muscular limbs should characterise the Shetland pony—long and powerful forearms and thighs, large, low-set knees and hocks, flat and clean bone below them, and fairly long pasterns. A common fault in some strains is lack of muscle in the second thighs, which is often so exaggerated that the pony has the appearance of having a deep hollow behind the thigh instead of an easy line from the quarters to the hocks.

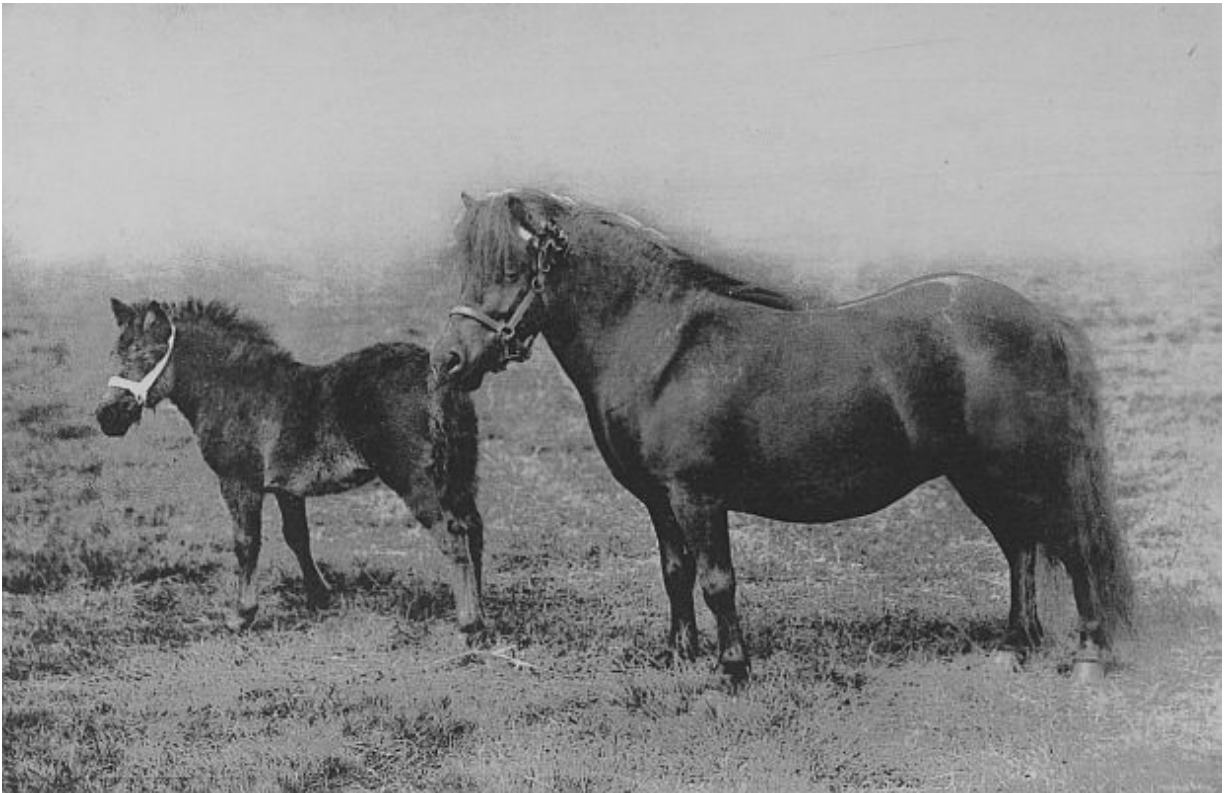
On the whole, the feet of the Shetland pony are good—large, round, and open, of fairly hard and very sound texture. Occasionally, however, narrow and contracted feet are found; and these should be regarded as a serious defect.

The coat of the pony is one of its most familiar and characteristic peculiarities, consisting as it does of fine thick fur below and an outer covering of longer and harder hair growing through it. Any weakness of coat is a serious fault, not only as being a departure from a deep-rooted characteristic of the breed, but also on the most practical grounds. No better protection could be imagined against wind and rain than the thick undercoat, waterproofed by the outer hair, from whose damped locks the water drips along the pony’s sides, while the under parts of the body remain dry and warm. In summer the shaggy coat,—falling off in ragged masses, is replaced by a sleek and fine hair. At all seasons the tail, mane, and forelock are as picturesque as they are useful in protecting the pony against weather and flies. They should be abundant, the hair strong in texture, and straight, falling flat, and, like the foot hair, free from any tendency to curl. It is an interesting characteristic of the Shetland pony that many animals shed in autumn the upper lock of the tail in such a way as to look as though the hair had been scrubbed off, although on actual examination it will be found that the hair is cast from the root and grows in again. A similar appearance is undoubtedly represented in prehistoric horse-portraits.

All colours are permissible in the Shetland pony, although black and dark-brown are now most common, and are preferred in the Islands by the

oldest traditions, which associate piebald and skewbald colours with softness of temper and with a strong suspicion of Norwegian cross. White markings in ponies other than piebalds and skewbalds are an undeniable blemish, particularly if they take the form of white stockings and the accompanying white hoofs. Dun, grey roan, and dappled grey are good colours, which should not be allowed to die out: the dappled grey should have blue hoofs.

Action is increasingly regarded, and rightly so, by judges in the showyard; and it is of the utmost importance in practice. It should, of course, be perfectly true and straight: dishing, straddling, and wide hock action are glaring faults. But action should also be vigorous, light, and springy, not showing the roundness that often disfigures hackney gait, but with fore-legs well thrown forward from the shoulder both in walking and trotting, while knees, pasterns, and hocks are freely and powerfully flexed. It must be admitted that in many Shetland ponies activity has been unduly sacrificed to abnormal shortness of limb. This is a point which demands careful attention; and it may be worth while to note that the ponies in the Islands are, as a rule, singularly active, as indeed the conditions of their existence require that they should be.



STELLA (1692.)

In one other respect modern show standards and conditions threaten rather to impair than to improve the breed. The appearance of the pony in the Islands almost invariably suggests a strong and vigorous frame: in the showyard there are few ponies whose appearance suggests any frame at all. This is, no doubt, greatly aggravated by the extreme and excessive fatness of most ponies in the ring; but it points to a real defect also. Every good horse ought to suggest to the imagination the general structure of his bony framework; and it ought scarcely to be possible to conceal this by any reasonable degree of condition, or to bring about, in a horse, a general appearance of bonelessness such as might be proper to the carcase of a perfectly fattened Aberdeen-Angus bullock. Partly from the practice of showing ponies much too fat, and partly also from the fact that breeders have neglected to seek for strength of frame as distinct from mere thickness of bones, the Shetland pony in the showyard has undergone some little deterioration in this respect. It should not be forgotten by breeders or judges

that a pony whose shoulders, hips, and stifles are not prominent in his appearance, is either defective in structure or very improperly overfed.

It may well be the case that a rather too exclusive use of the excellent “Londonderry” strain of ponies requires now to be corrected by a careful introduction of new blood, and that it is desirable to make use for that purpose of active, large-framed, vigorous mares of other Island strains. It cannot be too strongly emphasised that no risk should be run of sacrificing the results already attained—the power substance and well-formed joints and limbs, in which the “Londonderry” ponies excel. On the other hand, there is room, in very important respects, for progress and improvement in order to bring the Shetland pony to perfection.

CHAPTER IV.

The Management of Shetland Pony Herds.

ALMOST no other domestic animal is so easily cared for as the Shetland pony; and he appears to thrive and prosper in a bewildering variety of conditions. On the bare shores and hillsides of Shetland, where moss and seaweed must often supplement the scanty grass, he maintains as full a vigour, though perhaps not so comfortable an appearance, as he displays in the almost excessive luxury of some of the South-country studs.

It must not, therefore, be supposed that, in describing what appears to be the best system of management, we suggest that no less perfect system will satisfy the pony's needs. But, on the other hand, the best results can only be obtained by giving the most suitable conditions of development; and the fact that these conditions are not necessary to the life, or even to the health, of the pony, is no good reason for neglecting to provide them if they are really most favourable to its excellence.

Ponies will thrive on a great variety of pastures, and only careful management is needed to make any land serve their purpose; but it is of the greatest importance that they should not be kept always on the same ground. The need for a rotation or alternation of stock on pastures is generally admitted; and it is specially recognised in the case of horses, which are well known to make grazings "horse sick" if they are kept too constantly on the same fields. Shetland pony breeders are perhaps apt to be less careful in this respect than other owners of horses; and it cannot be too much insisted on that ponies ought to be kept in combination with cattle, either grazing with them or, preferably, alternating with them in the fields.

The quality of grass required naturally varies with the condition of the stock and the stage of its development. For in-foal and milking mares it is desirable to have fairly good pasture, such as would be suitable for dairy cows, so that mare and foal may be kept in good condition without over-exertion. On the other hand, the chief requirement of young ponies is abundant space; and large fields are much better for them than smaller

enclosures. They should, therefore, naturally be run out on much less luxuriant grazings than are desired for milking mares; they will thrive and develop well on such land as is suited for hill sheep; and, if this be not available, fields that have previously been closely grazed by other stock may be used to give them the space and freedom they require. In whatever way it may be obtained, this is one of the most important conditions of successful rearing; and breeders who have not land available to provide it would do well to board their young ponies out, in summer at all events, where they can find large stretches of clean grazing; for youngsters of a year old and upwards will maintain admirable condition in summer on poor land; and their activity and vitality are greatly increased by letting them find their living on pastures where the supplies of food are not too abundant, and where they are induced to travel constantly over fresh ground.

The foaling mares are the part of the herd to which most regular attention should be given. They must be kept in vigorous condition; and it must be remembered that they are under constant strain, giving large quantities of milk, and at the same time maintaining and developing the unborn foal. Horse owners are perhaps apt to forget that the mare is really a very large milk producer—certainly not less so than the average cow—and that provision should be made for this. It cannot be too clearly understood that this provision ought to be made in advance. Good pasture during the nursing season is, of course, desirable; but it is not sufficient, unless the mare is brought to the time of foaling in reasonably good condition. Foaling mares ought not, certainly, to be fat; but they ought to be in a robust and well-nourished state; and neglect of this must shorten their lives, both by general exhaustion and by accelerating, through loss of muscular tone, that “falling” of the womb which is the commonest cause of losses in foaling.

During the last few months, therefore, before the foaling season begins, the mares should be kept under observation, and supplied with hay if they seem to require it. Any which, from age or youth or other causes, are in specially poor condition, should be fed separately and receive perhaps some oats and bran.

So treated, the mares need usually give their owners no anxiety as foaling approaches. Care should, of course, be taken by selecting a proper date for mating, to have the foals born after the coldest weather of spring is likely to be over, and when some growth of grass may reasonably be expected. This

time will vary for different climates. In Scotland, the first days of June are usually the best time for mating, most of the foals being thus born during May, and therefore fit to be weaned in autumn.

It is the all but universal practice—and certainly the best and safest—to leave the mare entirely to her own devices during foaling. She should be left out in the field unless most unusual severity of weather prevents it; and in almost every case she will foal successfully without assistance,—indeed, when assistance is required it is very often unavailing. Mare and foal should be watched to make sure that the latter is sucking and is being allowed to do so; and in the rare cases in which any difficulty arises, help must be given by catching the mare and holding the foal to her. The only other danger that besets the young foal arises from a stoppage of the lower bowel which sometimes occurs. This is shown by the foal's frequent strainings, and can easily be removed by local action, when discovered; but neglect of it will result in the death of the foal in a very short time.

Most of the foals are likely to be born before the time for mating arrives; but, whether foaled or not, mares should all go together to the horse with which they are to run during summer. This is particularly necessary in the case of some horses, which, retaining the wild gregarious instinct, will not tolerate the addition of a new mare to the herd. If it be thought that the first ardours of the stallion are likely to disturb in-foal mares, or if he is suspected of any tendency, when excited, to attack foals (a possible though rare occurrence), he may be run for a few hours with fillies or barren mares, and the herd safely introduced when his excitement has subsided.

The Shetland pony herd is to be treated as a natural—practically a wild—herd of animals. The less the ponies are interfered with the better, so long as they have sufficient clean grazing and an ample water-supply. It is the experience of all breeders that the best results in the production of foals are obtained from running the horse constantly with the mares. The herd is kept together until a date early enough to avoid all risk of next year's foals being born too late for autumn weaning.

While proper care and management of the stock are essential to the best results, yet these results ultimately depend on the skill and judgment with which the breeding animals are selected and mated.

It is not proposed to attempt here to give rules for the exercise of the breeder's art. The principles of breeding are very much the same in every case. It is, above all, imperative—and especially in the selection of sires—to insist on soundness and vigour of constitution; and this becomes the more imperative the more we shelter the progeny of our stock from the rigour of natural selection, and from such severe tests of endurance as are imposed on race-horses. We have seen how closely inbred the leading families of Shetland ponies are; and, while it is wholly a mistake to suppose that this necessarily causes enfeeblement or unsoundness, yet it is an additional reason for exercising the greatest care in excluding these fatal taints.

In Shetland ponies also, as in other races of animals, the actual excellence of an individual is not a sufficient reason for expecting corresponding excellence in its progeny. Heredity is an element at least as important as good individual quality in the selection of sires and dams; and heredity itself—so complex are the elements that compose it—is a test of merit far less valuable and complete than the previous progeny of the animals to be bred from. Mating the best with the best, and breeding from long lines of fine pedigree, are both venerated rules; but the breeder who is fortunate enough to obtain animals already proved to be successful in their offspring has a surer ground than such rules give for expecting good results. It remains only that he should discover, by careful study and close consideration, with what type and heredity the animal he is about to breed from has been most usefully mated; and he may then hope to produce some proportion of stock approximating to the type he desires to embody. But he must, above all things, have a clear idea of what it is that he aims at creating or reproducing, not necessarily an idea to remain unaltered by experience and criticism, but yet a view and an aim independent of changes of fashion and of the varying fortunes of the showyard. Nothing but failure in breeding can result when a dominating purpose of this kind is absent.

The present-day breeder of Shetland ponies is neglecting to use the chief instrument ready to his hand if he fails to take great advantage of the admirable material created in and descended from the Londonderry Stud; and he ought specially to remember the value of the combination of Odin and Prince of Thule blood, which has already been referred to. But he ought not to make this his only source. The Islands still contain animals and

strains well fitted to be a strength to the breed; and one of the most interesting parts of a breeder's work consists in the careful and gradual introduction of these outside strains of blood.

The conclusion of the foaling period, and the completion of mating, open a peaceful and pleasant season in the pony-breeder's year—a season during which troubles and mishaps are usually few; while the contented mares, the antics of the foals, and the young stock in their summer bloom, form a picture contrasting sharply with other scenes in the passing of the year. The breaking up of the herds in August, or thereby, and the weaning of the foals in later autumn, bring this period to a close.

Weaning is a process requiring some little care and attention. The foals should be taken from their mothers not before the end of their fourth month, and preferably at least a month later; but weaning ought not to be unduly postponed, since it is important that the foals should have recovered from it before the severity of winter is felt. October is late enough for this, and late enough also to release from her nursing duties a mare which is to produce another foal in the following spring.



FOALS IN SUMMER.

The mares should be relieved, twice or oftener, of any severe pressure of milk after the foals are taken away, and be kept on poor grass for a day or two. The foals should be shut in until their first agitation is over, and be taught to feed from the trough. Any which may have been weaned earlier than is quite desirable are easily taught to drink separated cow's milk with some sugar added—the best of all substitutes for mare's milk. For the rest, there is no better food than bruised oats and bran, at first given as a mash and afterwards dry, with the addition of a small allowance of linseed meal, molassine meal, or molascuit. This feeding, with good hay and access to rough grazing, should be continued throughout winter. During this first winter liberal feeding is desirable; and adequate shelter should be given in the form of sheds or open loose-boxes, not to keep the foals warm, but merely to protect them from rough weather and to secure for them a dry lair in the long winter nights.

Older ponies need no such provision as this, though they are much the better of some such shelter as can be obtained from trees, dykes, good hedges, or steep banks. They should have ample grazing in fields left rough for the purpose, and should be supplied with hay when snow is on the ground, and at times when the winter grass proves insufficient for their needs. It ought to be kept in mind that stormy and wet weather are much more trying to them than hard frost or even snow, from both of which they seem to suffer little. Prolonged beating rain and damp ground to lie on tax their energies severely; and the wet and innutritious grass requires to be supplemented by dry food of some kind. In spring the rough pasture, which often seems to have been wasted in winter, repays its cost, for under its tufts fresh blades of grass spring early; and the ponies will be found eating old and new together, and showing the effect of the new growth in the slackening of their winter coats, which begin to fall off in large masses.

The period of weaning affords an opportunity of examining and treating the feet of mares and foals. The former usually require nothing but the shortening of the toes, and perhaps some paring of the hoof wall, with the removal of any inequalities of wear that present themselves. The foal's feet, however, often require a good deal of attention, specially in order to deal with cavities which are apt to be formed between the sole and wall of the hoof. These cracks or cavities should be freely opened up with the drawing-knife, explored and cleaned to the bottom, sometimes to a depth of over half an inch, and carefully packed with tow and tar. A similar examination should be made of all young ponies' feet twice a-year, and the teeth of aged ponies should also be carefully inspected.

An essential part of good herd management is the breaking of the ponies. No pony should remain unbroken; for, apart from every other reason, there is no means, other than breaking, for securing that combination of confidence and submission which every domestic animal should have. Every owner must have had experience of the inconvenience of having animals which cannot be handled without danger to themselves and their attendants, because they have never learned to yield to control, or to trust the ability and good intentions of man. With such animals ordinary management is difficult; and the treatment of illness, when it occurs, is hopelessly complicated.

But in addition to this sufficient practical reason, there is the further fact that without breaking there is no means of discovering whether an animal is or is not free from vice and ill-temper that make it undesirable as a sire or dam. It is unfortunately impossible to work all the Shetland ponies required to be bred from, although the ill effects of this are mitigated by the almost unvarying docility of the breed; but it is at all events desirable that the breeding stock should be tested for temper at some stage of their development.

Breaking is usually no difficult matter. A couple of lessons in leading, three in reins, and three in the shafts, with probably one severe conflict of wills in the whole process, will generally break a Shetland pony. A pony so broken is not of course a finished child's mount. Its mouth and manners are still to make; and they ought not to be neglected, for both can be perfect; and the pony's mouth particularly is naturally light and pleasant, although too often ruined by neglect and bad handling. All this should be carefully seen to when ponies are to be sent out to work; but for the purposes of herd management, the breaking just described is all that is needed.

Breaking is followed, in the case of show ponies, by preparation for show. The pony must learn to stand, walk, trot, and turn under such discipline as to present itself favourably to the judge. There is all the difference in the world between a pony showing his paces on a loose rein and going straight, true, and close, and one which must be held on a tight rein so that his head is turned round, his fore-feet almost forced to dish, and his hocks thrown out. The difference is sometimes one of temperament,—more often it is one of education. Training cannot turn a bad pony into a good one; but bad training may easily prevent the best of judges from seeing a good pony; and the fault is not with the judge but with the exhibitor.

The education of the show pony is a matter of time and patience—chiefly of endless patience. Some grooms have a genius for it, and those who have not must secure the result by greater labour; but in any case, careful practice and regular and sufficient exercise are the chief means by which the showyard results are obtained. As in every breed, preparation for show tends to be overdone. Over-fattening, as has already been said, is the most prevalent fault; but the employment of bearing-reins is sometimes carried far beyond the point required for that effective control which is the only

justification for tackle; and a prudent judge will never part from his work till he has seen the ponies without their trappings, and made sure that his selected winners can hold up their heads without the aid of straps. A more difficult problem lies in the tendency to the use of heavy shoes—a practice imported from the hackney stables to induce high action of the most useless and unsightly kind. A time may come when weights of shoes will have to be limited by rule; but it is to be hoped, rather, that firm and wise judging may convince exhibitors that true, sure, and clean action does not consist in the pounding motions produced by heavy shoeing. Good conditioning, development of muscle by exercise, and careful education are the legitimate preparation for show: everything else is a more or less successful attempt to deceive.

The diseases of the Shetland pony are comparatively few; but one or two are apt to occur even in well-managed herds.

Whenever the infection of strangles is brought in, it goes through the herd, attacking all the young animals and some even of the old.^[46] If it occurs in winter it may be necessary, in severe frost, to bring patients under cover to protect their wounds from frost-bite; otherwise it is best treated by keeping them out of doors; and so treated it is rarely a dangerous disease. Its symptoms are well known—the running at eyes and nose, the abscesses bursting as they mature, and perhaps forming and bursting a second time. Open air and liberal feeding are its sovereign remedies. In the case of foals a special difficulty arises from their inability to use the muscles of the swollen throat to suck, and this difficulty—dangerous if ignored—can best be met by milking the mare at frequent intervals and teaching the foal to drink from a pail, which it remains able to do.

Other troubles arise from worms—the common thread-worm and the deadly strongylus.^[47] In all cases of the latter veterinary advice must be obtained. But the best protections against these and similar troubles arising through infection are, first, to keep ponies always on the cleanest ground that can be given them; and, second, to supply them constantly, in every field, with rock-salt to fortify their blood and stimulate their digestion.

With these precautions and with ordinary care Shetland ponies give little trouble or anxiety so far as their health is concerned. The aim of herd management ought to be to supply the most natural life possible, so as to

reduce to a minimum the evils incidental to confinement within fences. This, with watchfulness and a due but not excessive liberality in feeding, will ensure health and long life to the ponies, and, to their owners, a reasonable profit and an unreasonable degree of pleasure.

CHAPTER V.

The Pony at Work.

It has already been explained that the Shetland pony is now little used for work in his native Islands, having been displaced by larger ponies and horses as the development of roads has substituted driving for riding, and carts for the creels and pack-saddles which are now found only in the remoter districts.



A TEAM OF MARES.

The progressive disuse, however, of the pony, as a work animal in Shetland, has been accompanied by a much greater increase in its use elsewhere. This has specially been the case in the employment of ponies in

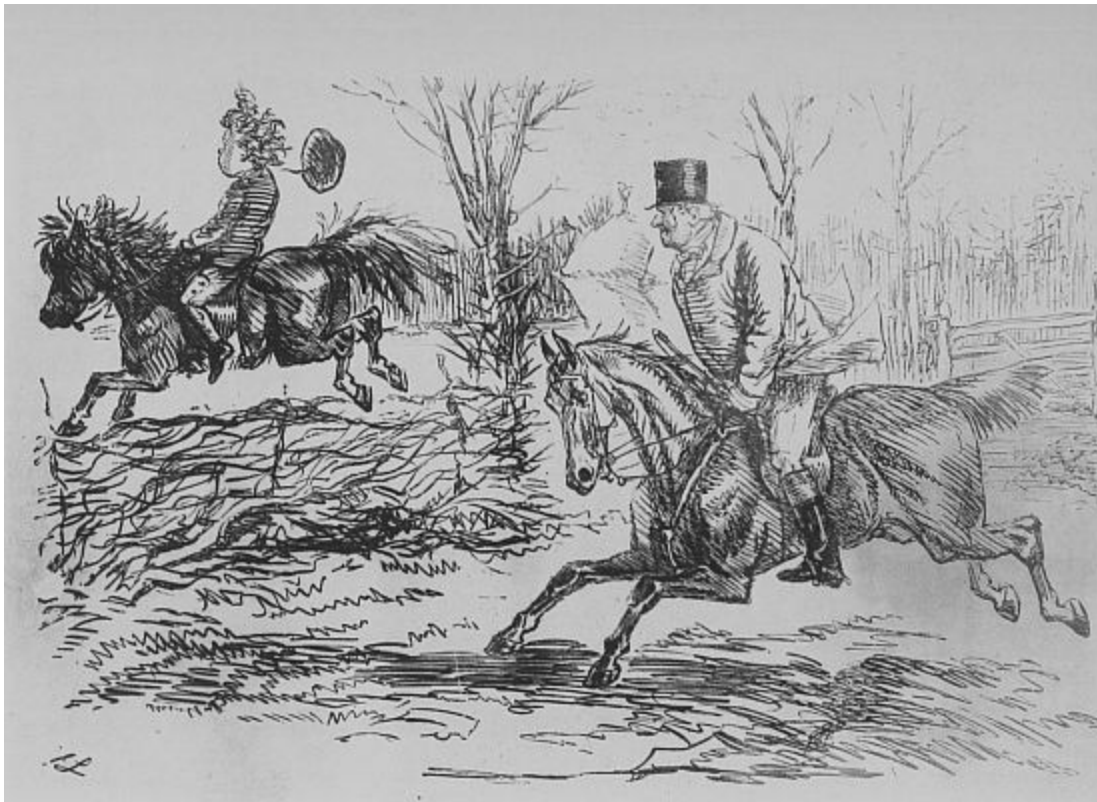
coal-pits, which grew to its height during the very period in which the use of roads became general in Shetland, and in which, therefore, but for some new demand for it, the pony might in all likelihood have come near to extinction.

It was in the middle of last century that ponies were first used in coal-pits in the north of England,^[48] horse ponies over three years old being imported at the price of £4, 10s. each. Their employment increased rapidly, until, in the late sixties and early seventies, the demand became so great that the earlier price was more than quadrupled; the supply of ponies was practically exhausted; and the Islands were all but depleted of good stallions; since these only are used in the pits so as to obviate the inconvenience of working stallions and mares together. The number now employed is very large, and shows little sign, so far, of reduction through the increased use of machinery, which tends to displace rather the larger horses used in the more spacious main roads than the Shetland ponies which can find their way in the lower-roofed passages of the pit. There is usually an active demand for thick, strong colts ready for work, which is not now permitted at less than four years of age. The Shetland pony—and particularly the pony of the “Londonderry” type—is admirably adapted for pit work. In structure he is exactly what is required, massive, muscular, and heavy, and yet able to walk comfortably in a passage not four feet high. His wise and placid disposition is no less a recommendation in an animal which is to work in cramped situations and in surroundings that might overstrain more excitable nerves than his. The Shetland pony learns easily to accept and adapt himself to new conditions; he travels with composure by sea or land; and the introduction of motor-cars into Shetland has been carried out with much less disturbance to the minds of the equine than of some of the human inhabitants: no animal could lend itself better to so strange a service as that of the coal-pits. Mr Brydon estimates that “it is not overstating the case to say that, on an average, they will travel over 3000 miles in the course of a year, and ‘shift’ as many tons of coal.”^[49]

A considerable sentimental repugnance exists to the employment of ponies below ground; and to the unaccustomed such a life appears sufficiently unattractive either to man or horse; but the fact remains that the life is not on the whole unhealthy: if it lacks the summer sun it is spared the winter nights. The fable that the ponies become blind has no better

foundation than is given by the fact that a short time is required to accustom their eyes to the light when they return to the surface. They are no doubt exposed to accidents; and they are less protected against overwork and other unfair treatment than they should be; though this is partly rectified by the provisions of the “*Mines Act*” of 1911.

It is only right to say plainly that the general accusation that has been made of cruel treatment of the ponies in the pits is an entirely unjust libel upon a class of men who, whatever their failings may be, are not inhumane. In point of fact, the ponies in the pit are usually sleek, fat, and contented, and display an affection for their attendants, and a confidence in man, not easily to be reconciled with the suggestion that they are habitually maltreated.



By permission of the Proprietors of 'Punch.'

A CHIP OF THE OLD BLOCK.

It must be admitted, nevertheless, that the pit-ponies are the less fortunate class of Shetlanders, and that the pleasanter career belongs to those which are selected for what appears to be the natural office of the pony—that of the child’s first mount. This is no new occupation for him. A letter, dated 1737, from “Mellerstains” to the writer’s brother in Bressay, runs: “In several of my letters I have told you that Lady Binning’s children are from home on account of their Education, so that a Horse would now be of no use as they’ll be grown up before they settle here again,”^[50]—a gift-horse, apparently, somewhat brusquely declined. Sixty years later a manuscript note, dated 1800, appended to a copy of Campbell’s *‘Political Survey of Great Britain’* (1774), bears that: “Yoked sometimes to the equipages of the Nobility, they have attracted the notice of ye metropolis.”^[51] We find the Sheltie, therefore, more than a hundred years ago in favour as a luxury, and nearly two hundred years ago recognised as a child’s pony. The pages of ‘Punch’ in the last century bear constant witness to its popularity for this purpose; and this could hardly be otherwise, for a child and a Shetland pony are an inevitable combination. Not that the pony requires so slight a burden; for it is not any ordinary weight of full-grown humanity, but only length of limb, that prevents the adult from riding it, as the Dutch seamen used to do in Shetland in spite of this drawback.^[52] But the pony attracts the child as no large horse can; and it is the ideal mount for early years. Its disposition is its first and greatest recommendation; for, while of course there are exceptions, generally the Shetland pony is so wise and kind and docile that it almost teaches the child to ride. Its manageable size and its admirable nearness to the ground promote the confidence which is the beginning of equestrian wisdom; and it never shatters or impairs that confidence by stumbling. This, indeed, is a qualification that is of capital importance. A pony that falls is of course an impossible mount for a child; but one that stumbles is scarcely better, since it constantly suggests the possibility of falling before experience and practice have neutralised fear. The breadth and strength and balance of a good Shetland pony make it the surest-footed of all riding animals. Theorists, without a fact to shelter themselves with, have alleged a danger to the health of children from sitting astride the big barrel of the pony, but the answer to them is quite simple. Their fears are imaginary, for they can produce no justifying instance; and anything less wide than the Shetland pony—in actual cross-measurement—would, owing to narrowness of chest, be an unsafe and stumbling mount: so long as men

ride, the Shetland pony will be the most valued and most valuable possession of the child happy enough to own it. But, to be the perfect child's hack or hunter, the Shetland pony must be bred as a riding-pony: it must have riding action—not the round and hammering gait of the once fashionable hackney, but the darting, gliding shoulder action that covers the ground quickly and smoothly. That alone is the safe and comfortable action of a saddle-horse large or small. Further, the pony must be bred with a short back and high withers to carry a saddle. The round low wither that disfigures too many modern ponies is fatal to the excellence of a saddle-horse, both because it will not hold a saddle in place and because it goes with short straight shoulders more proper to the coal-pit than to the road or field. Incidentally, it may be mentioned that a saddle, in the strict sense, is not necessary or desirable for a child's first riding. More ease and comfort—without any risk to the pony's back—can be obtained by using the treeless “pilch,” which pads the back and yet gives closer and safer contact than the saddle.

Trustworthy as the Shetland pony is, it must still be added that not only must he be educated, mouthed, and mannered as carefully as a larger horse, but also he should not be subjected to the temptations of power. Until a child can really manage and control him, the leading rein must be kept in regular use, so as to avoid those premature conflicts and accidents that are as fatal to future horsemanship as they are to equine manners.



ON DUTY.

The Shetland pony now goes far afield. In the United States he has enthusiastic supporters, who allow more laxity in height than British breeders approve—admitting 44 inches as a legitimate stature. He goes to Australia, New Zealand, the Argentine Republic, and South Africa, as well as to many European countries. In Canada, at the moment, he is in great demand: there he is the school pony; for in the new wheat lands farms are far from the schools, and a pony is the child's conveyance. For this purpose a mount is needed which is easily kept, docile, and hardy, and which can be hitched to a fence during school hours without being critical of the state of the thermometer. The Shetland pony supplies the demand, as if he had been created for that purpose; and Canadian buyers come to Britain year by year to take ponies in increasing numbers.

The Sheltie has the great advantage of a singular longevity. Every one who really associates with them knows how disastrously short a time dogs and horses live: on no reasonable calculation can they grow old with their

owners. Even the Shetland pony fails of this, but he makes the bravest of attempts. There are many accredited instances of ponies living to thirty-five years and upwards; while, among Stud-book ponies, the famous Jack died at the age of thirty, and his son Odin at twenty-four, while his grandson Thor still lives in health and vigour at the age of twenty-seven: with a little luck father and son may learn to ride on the same Sheltie.

The pony is the most easily kept of all animals. For two or three pounds a-year he can be maintained; for a little more he can be kept in hard-working condition—a useful member of a small establishment, and no unprofitable part of the equine staff of a farm, going over much more ground, with light loads, and a boy to drive him, than a cart-horse that will cost nearly ten times as much to keep.

Yet in the end it is idle to deny that it is not his indisputable economic validity that binds the Sheltie's lovers to him: rather it is himself—his wisdom and his courage, his companionable ways, his gay and willing service. Having taken from him their first falls and first riding lessons, and fought with him their first battles, they look forward to an old age in which he shall draw their bath-chairs; and in the interval of life he provides as a field animal the dual charm of a creature at once wild and tame—wild in his strong instincts, his hardihood, and his independence,—domestic in his wisdom and sweet temper, his friendly confidence in mankind, and his subtle powers of ingratiation.

APPENDIX

The Making of the Shetland Pony

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The Making of the Shetland Pony.

“The horses that are ancient we honour because we know not whence they came, but the new ones we slight because we know their beginning.”

THE Shetland pony belongs to an ancient breed famed for its intelligence and docility, strength and hardiness, but especially remarkable because of its small size.

In a recent article on the Shetland pony it is said that “the highest authorities rather incline to the view that he is an instance of arrested development, and that all the equine race originally sprang from ancestors far more diminutive than the smallest Shetland.”

It is doubtless true that the remote ancestors of the Equidæ were small, but it does not necessarily follow that Shetland, Java, and other pigmy breeds owe their diminutive size to arrested development. A human pigmy of West Africa is as well developed as a Hottentot of South Africa, and a toy terrier is as well developed as a mastiff. There is hence no *à priori* reason for assuming that Shetland ponies are not as well developed—mentally and physically as perfect—as Clydesdales or Arabs. Moreover, all animals during their development repeat, more or less, their ancestral history, climb their own ancestral tree, hence if there is arrested development we should find evidence of reversion to more or less ancient types. Is there any

evidence that in mind or body the Shetland is an instance of arrested development, or that he owes his diminutive size to reversion towards remote small ancestors?

It will be well at the outset to ascertain whether the small size is due to reversion or to dwarfing, induced, partly by unfavourable surroundings, partly by inbreeding and artificial selection.

The Size of the Shetland Pony.—Nature unaided has made a pigmy hippopotamus, pigmy elephants, and pigmy races of man, but there is no evidence that nature unaided in Europe or Asia in pre-glacial or post-glacial times produced a wild pigmy race of true horses—*i.e.*, of horses with only one complete toe for each foot.

The smallest wild horses in Britain at the end of the Palæolithic period (*i.e.*, according to a recent estimate some six thousand years ago) were apparently never under 12 hands at the withers. During the Bronze age, alike in wild and tame varieties, a size of at least 48 inches seems to have been maintained all over Europe. Further, remains from Roman military stations indicate that the smallest horses in Britain during the first century were probably never under 46 inches at the withers. It may hence be assumed that Shetland and other small breeds are not directly descended from pigmy wild races, but are the dwarfed descendants of one or more small varieties or breeds which had long lived under domestication.

A consideration of pigmy races makes it evident that dwarfing may be either equal or unequal, that it may result in the formation of a miniature having all the leading traits of the large race to which it belongs, or give rise to a pigmy variety in which certain parts are more dwarfed than others. In some small strains of dogs the relative proportion of all the parts are practically the same as in large strains, but sometimes in a small strain not only are the limbs more dwarfed than the trunk but certain parts of the limbs are more reduced than others. An example of unequal or disproportional dwarfing we have in the dachshund. In this breed the dwarfing has been carried further in the legs than in the body, and in the forearm than in the foot. In a normally constructed small hound in which the length of the body is 390 mm., the length from the elbow to the ground is 215 mm., from the elbow to the wrist 145 mm., and from the wrist to the end of the longest toe 95 mm. But in a typical dachshund with a body of

approximately the same size (390 mm.) the length from the elbow to the ground is only 137 mm., the distance from the elbow to the wrist being 95 mm. and from the wrist to the end of the longest toe 90 mm.—*i.e.*, in a dachshund, while the foot may only be reduced 5 mm., the reduction in the forearm may amount to 50 mm. (2 inches).

In the case of pigmy horses are the proportions of their normal ancestors invariably retained, or are the legs in some cases more dwarfed than the trunk, and as in the dachshund is the dwarfing greater in one part of the limb than in another? In Java ponies I have had under observation for some years the head and limbs bear practically the same relation to the body as in well-proportioned Arabs.

For example, in a 41-inch Java mare (fig. 1) the height at the withers, as in typical desert Arabs, is 2·7 times the length of the head, and the neck and limbs are relatively as long as in Arabs and other slender-limbed breeds.

But while in tropical islands the relative proportion of the various parts of pigmy horses may be maintained, in islands near the Arctic Circle dwarfing may imply undue shortening of the limbs, and that certain parts of the limb are more reduced than others.

A striking instance of unequal reduction we have in the Udganger or Nordlands ponies, once common in Bodo, a small island within the Arctic Circle off the coast of Norway. Fig. 2 shows that the limbs of the Bodo ponies were relatively nearly as much dwarfed as in a dachshund, while fig. 3 shows that Iceland ponies of the Nordlands type may closely agree in conformation with Exmoor and other well-built ponies of the Celtic race.

Very little is known about the make and size of the horses which first reached Shetland. The evidence as far as it goes indicates that they belonged to small varieties measuring from 11 to 12 hands at the withers. If horses were introduced from Norway during the Norse occupation, the majority of them would in all probability belong to the Nordlands race—*i.e.*, the race from which the modern fjordhest is believed to have mainly sprung. Probably unequal dwarfing more or less pronounced took place at a comparatively early period in some of the smaller islands, while in the more fertile parts of the main island, and in the rich island of Fetlar, the reduction in size (as in Java ponies) would be nearly uniform. It is conceivable that some of the unimproved ponies now living in Shetland, and also some of

the improved ponies bred and reared far from their ancestral island home, are as well proportioned as members of the Exmoor or Welsh breeds. One must, however, be prepared to find that not a few of the inbred pedigree ponies have undergone unequal dwarfing, one part of the limbs, as in the dachshund, having undergone more reduction than the adjacent parts.

Dwarfing in Shetland Ponies of the Celtic or Riding Type.—That well-proportioned Shetland ponies of the riding or Celtic type still exist is suggested by the measurements of Pamela and certain other fine-limbed pedigree ponies. Pamela (40 inches at the withers, 25 inches from elbow to ground, and 5·25 inches below the knee), in the form and length of the head, length of the limbs and their relation to the height at the withers, very closely agrees with the 41-inch Java pony.

The skeletons of Shetland ponies of the riding type available for study (viz., of Highland Chieftain, Egil, and Eric) also indicate that in a considerable number of cases the dwarfing is uniform. Though in many Shetland ponies the distance between the knees and the fetlocks looks very short, the front cannon-bones may be relatively as long as in thoroughbred race-horses. In Highland Chieftain^[F1] (fig. 4) the front cannon-bones (metacarpals) are 136 mm. long and 20 mm. wide at the middle of the shaft; in Persimmon,^[F2] the famous thoroughbred 16·2 race-horse (fig. 5), the metacarpals measure 276 mm. by 38 mm. As Highland Chieftain measured 33 inches, and has cannon-bones measuring 136 mm., he was half the height of Persimmon, and has cannon-bones practically half the length. In Highland Chieftain the cannon-bones (fig. 4) are not only relatively as long as in Persimmon (fig. 5), they bear almost exactly the same relation to the bones of the forearm and arm as in Persimmon—the *radius* (chief forearm bone) being relatively only 8 mm. shorter, the *humerus* (upper arm bone) relatively only 10 mm. longer. Nevertheless the cannon-bones of Highland Chieftain are relatively shorter than in typical Celtic ponies. In a 33-inch Shetland built on Celtic lines the cannon-bones should measure 142 mm., hence it may be assumed the cannon-bones of Highland Chieftain have been dwarfed to the extent of 6 mm. or one-quarter of an inch.

[F1] The skeleton of Highland Chieftain (a 33-inch Shetland pony bred in Scotland) is in the American Museum of Natural History, New York; Egil's is in the University of Edinburgh.

[F2] The skeleton of Persimmon is in the British Museum.

The only striking difference between Highland Chieftain and a typical 12-hands Celtic pony is in the face. In modern horses, while the form of the cranium or brain-box is nearly constant, the face varies both in size and in its relation to the cranium. In the wild steppe horse (*Equus przewalskii*) of Mongolia, which during part of the year subsists on hard dry food, the jaws are so long that the length of the face (fig. 8) is twice the width across the orbits, thus giving a frontal index of 50; whereas in a broad-browed Iceland or Highland pony the face may be only 1·6 times the width, which implies a frontal index of 60. In the Celtic race (to which Highland Chieftain mainly belongs) the normal frontal index is 54—*i.e.*, the length of the face is 1·8 times the width—but, as in Highland Chieftain, the length of the face is only 1·5 times the width, the frontal index^[F3] is as high as 65. Further, in Highland Chieftain the profile instead of being convex as in steppe horses (fig. 8), or nearly straight as in many Exmoor ponies, was decidedly concave or dished (fig. 4). The difference between the profile of a Shetland pony and that of a steppe horse is brought out by figs. 6 and 8.

[F3] The frontal index is obtained by multiplying the greatest width above the orbits by 100, and dividing the result by the length of the face, as measured from the alveolar point to a line connecting the supra-orbital foramina.

In the case of Egil^[F4] the dwarfing of the limbs seems to have been more pronounced. Egil may be regarded as an unimproved 40-inch pony; he met his death some forty years ago by falling over a cliff near Hillswick, Shetland—*i.e.*, before the Marquis of Londonderry set about making a short-limbed strain suitable for pit work. A comparison of the skeleton of Egil with that of an Exmoor pony in the British Museum shows that in the northern island pony the limbs were relatively shorter by one inch, and the face by half an inch, than in the southern moorland pony, but, notwithstanding the shortening of the limbs, the front cannon-bones in Egil, as in fine-limbed prehistoric races, are in length seven times the width.

[F4] Egil (a four-year-old black stallion) belonged to Mr Anderson, Hillswick.

As Egil probably belonged to an unimproved stock, it may be asked, Does his skeleton lend support to the view that modern pigmy horses reproduce, apart from their size, the characteristics of their remote ancestors?

In Miocene, as in prehistoric times, there were light as well as heavy horses, but in the light as well as in the heavy each limb had three hoofs (fig. 18). In *Neohipparion*, a late Miocene three-toed horse, about 40 inches at the withers, the skull is longer by 12 mm., and the molar teeth are more specialised than in Egil, the 40-inch modern island pony. In the Equidæ the cannon-bones are especially interesting; strange as it may appear, the middle cannon-bones are relatively longer and more slender in the extinct *Neohipparion* than in modern race-horses. In Egil the front cannon-bone is 175 mm. (6·75 inches) long and 25 mm. wide; in *Neohipparion* the corresponding bone has a length of 210 mm. (8·25 inches)—is longer than in a 48-inch Exmoor pony,—and is so slender that the length is nearly nine times the width—*i.e.*, relatively more slender than in desert Arabs. Further, in *Neohipparion* the middle metatarsal (hind cannon-bone) is as long as the femur (thigh-bone), whereas in even the 16·2 hands race-horse Persimmon (fig. 5) the middle metatarsal is only three-fourths the length of the femur. But, though in the 40-inch Shetland pony the skull and the cannon-bones are actually shorter than in the ancient 40-inch Miocene horse (*Neohipparion*), the second and fourth digits are as rudimentary, are as much “splint” bones, in Egil as in Arabs and thoroughbreds. There is hence no evidence of reversion in Egil, no attempt to reproduce the second and fourth toes, which, though shorter, were as complete in *Neohipparion* and his three-toed contemporaries (figs. 19 and 20) as the large functional middle toe,—in other words, in pigmy horses there is evidence of arrested growth but not of arrested development. In Egil, as in Highland Chieftain, there is also evidence of arrested growth in the facial part of the skull; the profile is concave and the frontal index high (65) as in Highland Chieftain.

Having seen that, apart from the face, the only essential difference between the skeleton of an unimproved Shetland and the skeleton of an Exmoor pony is a difference in size, let us next direct attention to the skeleton of Eric, a 36·5-inch improved pedigree pony of the riding type, for some time in the possession of Mr Charles M. Douglas of Auchlochan. In Eric, who died when six years old, the fore limb from the elbow to the

ground was 22 inches, the length from the point of the hock to the ground 15·4 inches, the circumference below the knee 5·25 inches, and the width of the fore-shank 1·2 inches. All four ergots were present, but the hind chestnuts were absent. Though the setting-on of the tail, the somewhat rounded hindquarters and the presence of ergots, indicated that Eric, like practically all modern Shelties, included horses of the “forest” type amongst his ancestors, his skull, teeth, and limbs made it evident that he mainly belonged to the Celtic or riding type.

In Eric the face (fig. 6) is so short that the frontal index is 67, the length being only 1·4 times the width instead of 1·8 as in 12-hands ponies of the riding type. The length of Eric’s head when alive was 410 mm. ($16\frac{1}{4}$ inches). A typical Celtic pony with a 410 mm. head measures 40 inches at the withers. Eric, though having the head of at least a 40-inch pony, only measured 36·5 inches. It may hence be assumed that, through dwarfing, his total height was reduced by 3·5 inches.

Further, as Eric’s metacarpal (fig. 9a), instead of measuring 166 mm. (the normal length in a 40-inch pony), had only a length of 143 mm., it follows that practically one inch of the dwarfing was due to a reduction in the length of the cannon-bones. Moreover, as Eric actually measured 36·5 inches, his metacarpals should have measured 152 mm. instead of 143 mm.—143 mm. being the length in a normal pony measuring 34 inches at the withers.

Though Eric had the head and trunk of a 40-inch pony and the metacarpals of a 34-inch pony, the metacarpals bear the same relation to the radius and the humerus as in Persimmon—*i.e.*, in Eric the relative lengths of the different parts of the limb were maintained (not lost, as in the dachshund) during the dwarfing process.

Although the cannon-bones in Eric had been considerably reduced in length, they had not been reduced in width—*i.e.*, they are as wide as the metacarpals of the 40-inch Shetland pony Egil, in which the limbs closely conform to the Celtic type. It thus appears that, in the case of ponies, reduction in height at the withers, and especially in the length below the knee, is not necessarily accompanied by loss of “bone.”

It was at one time the ambition of some breeders to have Shetland ponies as small as their remote three-toed Miocene ancestors. As a matter of fact,

ponies smaller than some of the Miocene species have long existed in Shetland. Eric, though an average-sized pony, was at least a hand smaller than the late Miocene horse, *Protohippus sejunctus*, in which the front cannon-bones (fig. 9b) were 177 mm. long and 21 mm. wide—i.e., 34 mm. (1·37 inches) longer but 5 mm. narrower than in Eric; while Seedpearl (31·75 inches at the withers) and other still smaller living ponies have shorter limbs than the very ancient three-toed *Mesohippus* from the Badlands of South Dakota. But while some Shetland ponies are actually smaller and have relatively decidedly shorter legs than the horses which flourished long before man appeared on the scene, they never have three toes and their teeth are always decidedly longer if not more complex than in the most advanced Miocene species. Hence, as already said, though in Shetland ponies there is evidence of arrested growth, there is no evidence of arrested development. It has been pointed out that the facial part of the skull of Eric is so short that the frontal index is extremely high—67 instead of 54. Even more remarkable than the shortening of the face is the reduction in Eric of the capacity of the nasal chambers. In new-born foals, owing to the relatively large size of the cranium, the face is always more or less dished (fig. 7). In the case of the wild horse of Mongolia, the increase in the size of the nasal chambers soon gets rid of the dishing, and in course of time the nasal bones are bulged outwards, so as to give rise to a more or less marked “Roman-nose” (fig. 8). But in Eric and many other Shelties of the riding or Celtic type, owing to the expansion of the nasal chambers being prematurely arrested, the profile in the adult (fig. 6) differs but little from that of the new-born foal (fig. 7).

Shetland Ponies of the Cart-horse or “Forest” Type.—For want of material nothing very definite can be said about the nature of the dwarfing of ponies of the heavy or cart-horse type. In a typical 12-hands Celtic pony the metacarpals are 200 mm. long, 27·5 mm. wide, but in a typical 12-hands “forest” horse the metacarpals are, on an average, 193 mm. long and 35·1 mm. wide. An undwarfed, thick-set, 36-inch Sheltie built on the same lines as a 12-hands “forest” horse should have metacarpals about 145 mm. in length and 26·4 mm. in width, and should measure 5·5 inches below the knee.

The measurements available indicate that in a pedigree 36-inch Sheltie of the “forest” type, the front cannon-bones will probably measure 137 mm. by 26·2 mm., that the circumference below the knee will be 5·5 inches, and the distance from the elbow to the ground 21·5 inches—in Eric (36·5 inches) the length from elbow to the ground was 22·25 inches. If these figures are approximately correct, it follows that in a 36-inch Sheltie of the cart-horse type the limbs may be at least an inch shorter than in a dwarfed 36-inch pony of the riding type, and that the dwarfing may be unaccompanied by any loss of “bone.”

This conclusion is supported by the measurements of Odin, a 38-inch pony, 6 inches below the knee; of Vulcan, a 32-inch pony, 5 inches below the knee; and of other ponies of the Londonderry type belonging to the Ladies Hope, and also by those of Everlasting, Frederick, and other thick-set Auchlochan ponies. For example, in Everlasting, a 38-inch pony, the distance from the elbow to the ground is 22·75 inches, the circumference below the knee 5·5 inches, the bone being “round,” and the shank ·5 inches broader than in the flat-boned riding pony Eric.

In heavy horses, but especially in Shire colts, one (or more) of the limbs has occasionally an extra digit ending in a well-formed hoof. In a three-weeks’ horse embryo there are no rudiments of limbs; at four weeks the limbs are represented by paddle-like structures; at five weeks the paddles contain rudiments of three toes—miniatures of the three toes of *Hypohippus* (fig. 20).

In ordinary circumstances the development of the outer and inner toes (ii. and iv.) is soon arrested, but occasionally one of these rudiments develops into a toe as large and complete as in three-toed Miocene horses of the “forest” type. When this happens, when in addition to the third toe there is a toe corresponding to the human forefinger, we have a marvellous instance of reversion.

If in the Shetland breed there is a tendency to reversion, one would expect to find now and then extra digits in ponies of the heavy or “forest” type. I have, however, never heard of a Shetland pony with extra digits.

Dwarfing of the face and the reduction of the nasal chambers has apparently been carried further in some of the miniature cart-horses than in Eric and other flat-boned Shelties. In Jupiter,^[E5] e.g., the head, though

wider across the orbits, is shorter than in Eric, and decidedly more dished. Ancient horses adapted for a forest life had face pits in front of the orbits, which probably, like the corresponding pits in deer, lodged scent glands. Further, in ancient “forest” horses the upper lip was probably decidedly longer and more prehensile than in modern breeds. In broad-browed, big-boned Shetland ponies there is no indication of a pit for a scent gland, but there is sometimes an unusually long and decidedly mobile upper lip, which may or may not be due to reversion.

[E5] Jupiter was a 37·5 inch “elk-nosed” pony, with a girth of 54·5 inches, rounded quarters, a low set-on tail, a complete set of chestnuts, wide open hoofs, and six inches of “bone.”

The Causes of the Dwarfing of Shetland Ponies.—Given sufficient food and shelter, horses up to 15 hands—the size of the tallest prehistoric Old World wild horse—can easily be bred and reared in both the Western and Northern Islands of Scotland. On the other hand, large breeds are soon dwarfed when, in addition to a limited supply of food, the conditions during a considerable part of the year are extremely unfavourable. If Shetland ponies have not sprung from a small wild pigmy race, it may be safely asserted that their small size is mainly due to isolation in small areas where they were forced to shift for themselves under, as a rule, extremely unfavourable conditions. [E6] Obviously the environment may play a double part. It may (1) arrest growth by failing to provide sufficient food and shelter, and (2) it may eliminate the individuals which, by growing beyond a certain size, require during times of stress more food and shelter than are available. Considerable stress has been laid by writers on the dwarfing influence of the surroundings. It is said, *e.g.*, that “horses taken to the barren and cold islands of Shetland become gradually smaller and hardier, like ponies, and the hair becomes thicker and longer. Long-continued exposure to such conditions ultimately results in the production of an animal like the Shetland pony, small in size, extremely hardy, able to withstand the most severe winter climate, and to subsist on a minimum of food.” [E7] It might be said that this view is supported by experiments in the Western Islands. At the beginning of the eighteenth century the Captain of Clanranald (who was killed in 1715 at the battle of Sheriffmuir) brought from Spain “some Spanish horses which he settled in his principal island of

South Uist. These in a considerable degree altered and improved the horses in that and the adjacent islands. Even in 1764, not only the form but the cool fearless temper of the Spanish horse could be discerned in the horses of that island.... These at that time, both in figure and disposition, were thought the best horses observed in the Highlands, and though of low stature were judged more valuable than any other horses of the same size.” The descendants of the Spanish horses introduced by Clanranald for a time increased the size of the horses in the adjacent islands. Nevertheless, in spite of the introduction of the Spanish horses at the beginning of the eighteenth century, and the introduction of many south-country horses during the second half of the eighteenth century, the Highland and Island horses at the beginning of the nineteenth century were “sometimes only 9 and seldom 12 hands high.” Moreover, though some of them were “of an excellent form,” with “great strength in proportion to their size,” agile and spirited, many were “short necked, chubby headed, and thick and flat at the withers.”^[F8]

^[F6] That the conditions are now and again very trying in Shetland is proved by the death-rate among native sheep being, in some districts, from 40 to 50 per cent. during the winter of 1912–13.

^[F7] ‘Cyclopædia of American Agriculture,’ vol. iii. p. 34.

^[F8] Walker’s ‘History of the Hebrides,’ vol. ii. 1808.

Evidence in support of the view that the descendants of the Spanish and south-country horses, introduced about 1710 and after 1745, were soon either eliminated or dwarfed, we have from Dr Johnson and others. Dr Johnson tells us that the pony he rode in Coll was very low, and that a bulky man on one of the native horses made “a very disproportionate appearance,” and, after referring to the small horses of Rum, mentions that he had heard of a yet smaller race in Barra, “of which the highest is not above 36 inches.” Barra was one of the islands which benefited by the introduction of Spanish horses by Clanranald.

That dwarfing may be the direct result of an inadequate supply of nourishment during development is suggested by the condition of fœtuses I

found some years ago in a wild rabbit. The right uterine horn of this rabbit contained four young, the left eight. The eight in the left horn were as well developed as the four in the right horn, but only half the size and half the weight. Evidently the amount of nourishment available in this case was limited, and as the eight in the left horn only received as much as the four in the right horn, they were only half the size.

But while dwarfing may be, or appear to be, directly caused by the environment, there are other possible causes. Sometimes the small size of one or more members of a family or litter is due to reversion to small ancestors. For example, in a litter of five puppies bred some months ago (from parents in which West Highland and Mexican (Spanish?) blood prevails) there is marked variation in size. When these pups of mixed origin were weighed three days after birth, two males weighed 7·5 oz. each, two females 5·5 oz. each, and one male 4·5 oz. When again weighed at the end of the twelfth week, the largest male scaled 106 oz., the smallest male 44 oz., the larger female 58 oz., and the smaller female 46 oz. In this case the small male reproduced his small Mexican great-grandsire, while the large males took after their West Highland ancestors.

But dwarfs often enough turn up in old-established “pure” breeds, and now and again a dwarf is found in an otherwise normal human family. There is no reason for supposing that such dwarfs are the result of reversion. Just as one of a litter of pups may prove a dwarf, one of five or six foals, full brothers and sisters, may prove a dwarf.

It may, I think, be assumed that in the case of horses living under natural conditions, “spontaneous variation,” without the aid of reversion, will, as a rule, provide sufficient material to admit of a variety being evolved well adapted in size and other respects for the conditions which at the moment prevail.

It need hardly be pointed out that little will be gained by speculating as to whether dwarfing is due to the direct influence of the environment, to reversion, or to spontaneous variation.

Many breeders, more especially breeders of dogs, seem to think that dwarfing is invariably the result of inbreeding. It is doubtless true that the members of many recently formed pigmy breeds are closely inbred, but it is well to bear in mind that there are closely inbred large as well as closely

inbred small breeds—that size is largely a matter of selection—that in the case of natural races size depends more on the surroundings and the extent of the range than on the consanguinity.

The view that dwarfing is caused by inbreeding is insisted on by Sir Everett Millais in a book on Rational Breeding. Sir Everett states that, though in the case of the Shetland pony the “climate, bad food, &c., had been a factor in reducing the size, the primary cause was inbreeding due to isolation.”

It seems to me, however, highly probable that until artificial selection began in earnest about thirty years ago, inbreeding had little influence in determining the size of Shetland ponies—that isolation had been a decidedly more potent factor than inbreeding. Scottish red deer are decidedly smaller now than they were in Roman times; but this is not so much due to inbreeding as to the range of most of the herds being restricted, and to the best stags being cut off before they have a chance of improving the herd. The Scottish deer in New Zealand, though all descended from a few imported individuals, instead of dwindling in size, are larger and carry finer heads than their home-bred relatives,—the wider range and better conditions have more than compensated for the inevitable in-and-in breeding. It is doubtless true that in-and-in breeding sooner or later diminishes the vigour, size, and fertility, and, in addition, restricts variation. But under natural conditions, if the range is sufficiently extensive, occasional reversion to vigorous ancestors will prevent dwarfing provided there is rigid elimination of the unfit.

If Shetland ponies are the pigmy descendants of one or more ancient races at least as tall as Exmoor and Welsh ponies, one would expect them to increase in size when bred and reared under favourable conditions. It has been again and again asserted that “the climate and comparative privation of the Shetland Isles were necessary to maintain the small stature of the ponies, and that the breed would inevitably lose this and all other characteristics if bred away from Shetland and under more generous conditions.”

Some of the ponies recently brought south from Shetland have increased so much in size that, if otherwise eligible, they could not be registered in the ‘*Shetland Pony Stud-Book*’—i.e., they are now over 42 inches at the

withers.^[F9] The majority of these tall ponies, however, are piebalds or skewbalds, which in make and other respects resemble Iceland ponies,—if their history were traced, a piebald Iceland pony would probably be found amongst their recent ancestors. It is not surprising that young cross-bred Shetland ponies increase considerably in size when grazed on rich lowland pastures, or that now and again a pure-bred pedigree pony should grow above the recognised standard; but these exceptions only serve to prove the rule, now widely recognised by breeders, that pure-bred Shetlanders remain small however favourable their surroundings. Seeing that in the majority of pedigree ponies the dwarfing has gone so far that the metacarpals are actually shorter than in the remote three-toed Miocene ancestors, what is surprising is that there is not an immediate response to the stimuli which genial surroundings and abundant food imply.

^[F9] In all probability these ponies would have measured over 40 inches had they remained in Shetland.

Some 500 years ago a female rabbit and her young were turned out on the small island of Porto Santo near Madeira. In course of time this island was so overrun with rabbits that it was for a time abandoned as a settlement. As the rabbits increased in number they dwindled in size, became reddish above and grey beneath, and lost the black marking from the points of the ears and the tip of the tail. Some of these Porto Santo rabbits which reached the London Zoological Gardens in 1861 reacquired the colour and markings of the common wild rabbit within four years. The Porto Santo rabbits having recovered the ancestral colours soon after reaching Europe, it might have been anticipated that the Shetland pony would recover some of his lost inches when taken to the south of England—to the area containing the remains of the 12-to 13-hands wild races from which small British breeds have mainly sprung. There is, however, no longer any doubt that the small size is, as a rule, maintained however favourable the surroundings. Why the Sheltie fails to respond to the growth stimuli which favourable surroundings so abundantly provide it is extremely difficult to explain. In the language of the day, one can only say that the limbs have forgotten how to grow beyond a certain size, and add that this loss of “memory” may be the result of breeding from the smallest individuals regardless of their consanguinity. In 1892 Mr Christopher Wilson, in a letter to Mr Meiklejohn, then in charge of

the Bressay Stud, said: "With regard to the Shetland ponies, your great object is to keep them small.... There is no class of animal to which inbreeding can be better applied, as all you will lose by inbreeding is size, which is what you want.... To inbreed them there is only one plan. Select your very best stallion and put him to a certain number of mares, and then put all the good fillies when three years old to their father; also, select your very best mare and put her to another stallion, and go on breeding from her to the same stallion until you have a colt, then put that colt to his mother, and use the produce to breed from with the produce obtained by putting the fillies above mentioned to their father." A better plan for fixing the size could hardly be imagined. If followed for some years the size of the bones and muscles would doubtless be so effectively stereotyped that, however much their "memory" was jogged by "fresh fields and pastures new," there would be little or no response. Although close in-and-in breeding was undoubtedly practised, the Shetland Pony Stud-Book indicates that the advice of the originator of the "Sir George" strain of hackney ponies was not literally followed.

The Ancestors of the Shetland Pony.—From pigmy horses it seems but a step to the little "fossil" horses of bygone days. It is hence not surprising that many are led to inquire to which particular branch of the equine family tree the Shetland pony belongs.

Not many years ago it was said the horse tribe was at the start represented by a primitive race about the size of a fox but with as many hoofs as a tapir—four in front and three behind,—that in course of time a new race arose (with three hoofs in front as well as behind) which eventually gave rise to the one-toed ancestor of all the living Equidæ.

Now, however, we know that in Eocene times there were several kinds or species of four-toed "horses" from which were derived, in the Oligocene and Miocene periods, numerous three-toed species, some of them doomed to early extinction, others to bring forth one-toed races which in due time produced the ancestors of the modern horses, asses, and zebras.

Hence it is now admitted that, in the past as in the present, the horse tribe was always represented by several species, not a few of which at times

occupied the same area. It is also now admitted that the modern domestic breeds include several wild prehistoric species among their ancestors.

Is it possible to say which of the wild horses of prehistoric times contributed to the making of the Shetland pony? To be in a position to hazard an answer to this question it will be necessary to refer to the more important links which are believed to connect modern breeds with *Hyracotherium*, generally regarded as the remote polydactylous ancestor of the horse family.

Hyracotherium lived in the south-eastern part of England—his remains occur in the London Clay near Herne Bay and in the Red Crag of Suffolk—*i.e.*, in deposits formed two or three or it may be six million years ago. This ancient fossil horse—though not unlike a long-faced fox terrier—Owen regarded as a relation of the Hyrax or Coney of Palestine, hence the name *Hyracotherium*. Evolved in Western Europe—perhaps in England—this primitive small-brained terrier-on-hoofs not only wandered across Europe and Asia, but actually crossed into America—then freely connected with north-eastern Asia—and ranged at least as far south as New Mexico. So conservative was this small forerunner of the great horse family that the American representative from the Wasatch deposits only differs from the parent form in having slightly more complex teeth. The fore and hind foot of the American variety of *Hyracotherium*—generally known as *Eohippus*—is represented in fig. 10, and fig. 12 gives a restoration of *Eohippus*.

Though at the beginning of the Eocene period the milk-givers were backward and small (*Eohippus* was probably only 12 inches at the withers), the struggle for existence was probably keen enough. At any rate, the European varieties of *Hyracotherium* either died off without leaving descendants or gave rise to odd-toed ungulates which took no part in forming the modern Equidæ. But for the American branch of the *Hyracotherium* family there would have been no horses. In course of time *Eohippus* was supplanted by *Protorohippus* (a 14-inch horse longer in the limbs and with more complex teeth), which towards the close of the Eocene period gave place to the still more specialised *Orohippus*. It may be mentioned that *Epihippus*, a slightly modified descendant of *Orohippus*, may have lived side by side with the remote ancestor of the camels—a quaint even-toed ungulate about the size of a “jack-rabbit.”

The limbs of *Orohippus*^[F10] are represented in fig. 11, and a restoration is given in fig. 13.

[F10] In the Yale collection there are five species of *Orohippus* from New Mexico and Wyoming.

[F11] R. S. Lull, 'American Journ. of Science.' 1907.

North America during Eocene times “was clad with forests in which grew both evergreen and deciduous trees distinctly modern in character. The moist climate gave rise to many streams and lakes, along the shores of which grew sedgy meadows that in turn gave rise to grassy plains.”^[F11] In the following (Oligocene) period similar conditions for a time prevailed, but later, owing to the increasing aridity, broad meadow-lands and prairies made their appearance. The new environment produced larger and more active flesh-eaters, fleetier and more intelligent horses. One of the new and improved species is *Mesohippus bairdi*, an 18-inch horse with only a splint-like metacarpal representing the outer or fifth digit—a digit complete in all the Eocene horses. In this as in all the other Oligocene horses three of the four premolars, as in the recent Equidæ, resembled molars. Small and slender-limbed, *Mesohippus bairdi* was adapted for living in the open, but a larger species (*Mesohippus intermedius*) might be described as a “forest” horse,—though only 24 inches high, this forest-dwelling form had as long cannon-bones as a 33-inch Shetland pony of the “forest” or cart-horse type. Another American Oligocene type (*Miohippus*) from Oregon deserves mention, not so much because it was more specialised, but because it had a representative (probably a descendant) in Europe known as *Anchitherium*, which in Miocene times ranged from France to Bavaria. In this European species the last vestiges of the first and fifth digits had apparently disappeared. The fore limb of *Mesohippus* is represented in fig. 18, and a restoration is given in fig. 14.

It is impossible to say how many thousands of years are represented by the Eocene and Oligocene deposits, but an idea of the time that has elapsed since the beginning of the Tertiary period will be gained if it is mentioned that “when the fox-like *Hyracotherium* was wandering on the marshes of Kent not only was the Himalaya non-existent, but that along the line of its

very heart—where the kiang now lives at an elevation of from thirteen thousand to sixteen thousand feet—extended an arm of the sea of no inconsiderable depth.”^[F12]

^[F12] Lydekker, ‘The Horse and its Relations,’ p. 242.

During the Miocene period the horse passed through the most interesting phases of his evolution, his elaborate dental battery was almost brought to perfection, and the second and fourth toes were gradually dwarfed and hidden out of sight, not even a trace of the hoofs being left, as in sheep, to suggest polydactylous ancestors. As already hinted, though Europe was the birthplace of the remote ancestor of the Equidæ, it is in the Miocene deposits of North America that we have a record of the most important phases of their evolution. The remarkable progress made in Miocene times was doubtless necessary to enable horses and other grass-eaters to keep abreast of the profound changes in the environment—the great increase of prairies in some areas, and the upheavals which resulted in the appearance of extensive mountain ranges in others.

The Oligocene species which proved sufficiently plastic to respond to the new conditions varied in different directions, and gave rise to, amongst other types, one well adapted for a forest life, and one highly specialised for ranging far and wide over boundless prairies.

In *Hypohippus* we have an example of a “forest” horse, in the American *Hipparion* (*Neohipparion*) we have a horse more specialised for a desert life than the fleetest Arab, while in *Merychippus* we have a link with Oligocene species deserving attention, because, on the one hand, it gave rise through *Neohipparion* to the *Hipparions*, now extinct, but once common in Europe and Asia; and because, on the other hand, through *Protohippus* it seems to be the ancestor of the slender-limbed species of the “desert” or plateau type, now best represented by Celtic and Mexican ponies.

In *Merychippus* the orbit is complete, and the crowns of the permanent molars are cemented as in recent horses, but the hoof bone has a cleft at the apex (fig. 19), and in some cases there is a minute vestige of the “splint” bone of the fifth or outer digit of the fore-foot. In fig. 11a, the fore-foot of

Neohipparion, the Miocene race-horse, is represented. Fig. 19 gives the fore-foot, and fig. 16 is a restoration of *Merychippus*.

Hitherto *Merychippus*, through *Protohippus* and *Pliohippus*, has been by many regarded as the progenitor of all the modern horses, as well as of the extinct *Hipparions*. That slender-limbed horses with short-pillared molars are mainly descended from one or more varieties of *Merychippus* is possible, but it seems to me that modern breeds with short broad cannon-bones and long-pillared molars are probably mainly descended from browsing ancestors with limbs of the *Hypohippus* type.

Hypohippus, like *Eohippus*, but unlike all the known Oligocene horses, had in the fore limb, in addition to three complete and functional toes, a distinct vestige of the first metacarpal—*i.e.*, of the bone which in man carries the thumb. No vestige of a first metacarpal has ever been found in slender-limbed breeds, but once and again a vestige of the first metacarpal occurs in coarse-limbed breeds. The vestigial first metacarpal, taken along with other facts, suggests, as already said, that coarse-limbed breeds include a browsing race with limbs of the *Hypohippus* type amongst their ancestors. A few years ago it was assumed that *Hypohippus*, the 40-inch forest horse of Dakota and Montana, became “extinct during the Miocene, leaving no descendants.”^[F13] Now, however, it is admitted that browsing horses possibly “identical with *Hypohippus* of the Miocene of America”^[F14] lived in China at the beginning of the Pliocene. Though it is inconceivable that a species with the short-crowned teeth of *Hypohippus* could give rise to any of our modern breeds, it is possible that in the Pliocene of Eastern Asia a race (with *Hypohippus*-like limbs but long-crowned molars) may be found bearing the same relation to long, low, big-boned modern “forest” horses that *Merychippus* bears to fine-boned modern plateau or desert horses.

^[F13] Lull, *loc. cit.*, p. 177.

^[F14] Osborn, ‘Age of Mammals,’ p. 333.

The fore-foot of *Hypohippus* is represented in fig. 20. When a toe corresponding to the second digit of *Hypohippus* (II. fig. 20) appears in a modern horse it has sometimes, as in *Hypohippus* and *Eohippus* (I. fig. 10),

a vestige of the first metacarpal at its upper. Fig. 15 is a restoration of *Hypohippus*.

Ponies in Prehistoric Times.—During the Pleistocene period some eight or more species of true horses and ponies inhabited North America. Apparently before Palæolithic man reached the New World all these American species had become extinct. About the American true horses which, like *Hipparion*, reached and found a home in Eastern Asia, very little is known. Some of their descendants found their way during Pliocene times into India; others reached south-eastern Europe.

One of the Indian Pleistocene species (*E. namadicus*) from the Narbada valley had large long-pillared molars like *E. complicatus* of North America and *E. fossilis* of England (fig. 22); another (*E. sivalensis*), well represented in Pliocene deposits of the Indian Siwaliks, is the oldest true horse about which we have definite information. Like *Pliohippus*, *E. sivalensis* had short-pillared molars, but instead of measuring, like *Pliohippus*, 12 hands at the withers, this Indian species reached, in some cases, a height of 15 hands.

Some of the Kirghiz breeds, in which the face is strongly bent downwards on the cranium, probably include this ancient Siwalik race amongst their ancestors. Some of the Eastern races which reached Europe^[F15] in pre-glacial times found a congenial home in Tuscany and Umbria. Others, moving in a north-western direction, found their way into Britain, while others crossed by one or more land connections into North Africa. About the late Pleistocene descendants of the varieties and species which reached Europe before the Ice Age, a considerable amount of information has been gained from engravings and coloured drawings on the walls of caves occupied by Palæolithic man, and from fragments of skulls, teeth, and limb-bones found in Pleistocene deposits. Up to the end of last century naturalists, as a rule, assumed that the wild horses hunted during the Early Stone Age all belonged to the same species, the so-called (*E. fossilis*), and when about 1870 Przevalsky discovered wild horses in Mongolia, it was further assumed that these wild herds were the descendants of *E. fossilis*, and hence represented the wild species from which all the modern domesticated breeds had sprung.

[F15] The horses which reached Europe in Pliocene times are usually said to belong to one species (*E. stenonis*).

Partly from fossil teeth and limb-bones, and partly from the engravings on the walls of caves and on pieces of horn, the conclusion was arrived at that *E. fossilis*, the assumed common ancestor of modern breeds, was characterised by a large coarse head, coarse limbs, and long-pillared molars (fig. 22). Prjevalsky's horse when first discovered was said to be characterised by coarse limbs as well as by a large heavy head. As it was further assumed a generation ago that all domestic horses had long-pillared molars, and that the cannon-bones varied with the surroundings—being short and broad in some areas, long and narrow in others—there seemed no escape from the conclusion that all the horses now living under domestication are descended from one and the same wild Pleistocene ancestor.

But though in Prjevalsky's horse the head is coarse, the limbs are nearly as fine as in thoroughbred race-horses, and though in horses of the Prjevalsky or steppe type the pillars of the molars are long, they are not long in all modern horses. Moreover, though there is evidence of the existence in Europe in prehistoric times of a species with a coarse head and relatively fine cannon-bones, there is no evidence of the existence of a species which combined a coarse head with short broad cannon-bones. On the other hand, it has been ascertained that since Miocene times there have been living side by side in Europe big-boned and fine-boned species, and that, except by dwarfing, long narrow cannon-bones are rarely if ever transformed into short, broad cannon-bones.

The skulls from the Roman military station at Newstead proved conclusively that there lived in Scotland during the first century large and small horses with *short-pillared* molars (fig. 23). This led to the discovery that in Shetland and other ponies of the Celtic type, and in Arabs and thoroughbreds of the Libyan or Siwalik type, some of the molars have as short pillars as in *E. stenonis* (fig. 21) of the Val d'Arno Pliocene deposits.

The investigations of the last decade having indicated that during the Ice Age in Europe, as in America, there were always several species of horses living contemporaneously, an attempt must be made to ascertain from which of the wild prehistoric species Shetland ponies are mainly descended.

In addition to steppe horses of the Prjevalsky make and horses allied in skull, teeth, and limbs to *E. sivalensis* of India, there were in prehistoric times large and small varieties of browsing or forest horses, and “Celtic” and “Libyan” varieties of slender-limbed plateau or desert horses. Although trees and plateaus are conspicuous by their absence in the northern islands, the study of Shetland ponies makes it evident that they have mainly sprung from “forest” and “plateau” ancestors. Evidence of this we have in the limbs and skull as well as in the teeth. A fairly accurate estimate of the type to which a horse belongs, and also of its height, can often be gained from a study of the cannon-bones. For example, in 12-hands ponies of the “Celtic” variety the metacarpals are on an average 200 mm. long and 26·6 mm. wide at the middle of the shaft, whereas in 12-hands ponies of the “forest” type the length is on an average 193 mm. and the width 35 mm.—*i.e.*, in the one case the length is 7·5 times the width, in the other only 5·5 times. In Preglacial times there were horses in Umbria with metacarpals 190 mm. by 32 mm., and horses with metacarpals 220 mm. by 30 mm. The 190 mm. metacarpals probably belonged to a long, low, broad-browed 12-hands “forest” pony, while the 220 mm. metacarpals doubtless belonged to a 13-hands fine-limbed pony of the “desert” type. During the Glacial period horses of the “forest” and “desert” as well as of the “steppe” type were common all over Europe.^[F16] From the “Elephant-Bed” at Brighton bones of a 12-hands “forest” horse have been recovered. Kent’s Cavern, Torquay, in addition to the bones of a 12-hands “forest” horse, has produced cannon-bones of the same size and width as the fine-boned 13-hands Umbrian pony.

^[F16] In the vicinity of an open-air settlement to the north of Lyons there are rubbish-heaps said to contain the remains of over 50,000 horses, which served as food during the Solutrian period of the Stone Age.

The “Elephant-Bed” and Kent’s Cavern small “forest” horses are best represented to-day by the long, low, broad-browed Iceland ponies, while the 13-hands small fine-boned race of Kent’s Cavern is best represented by Exmoor and other ponies of the “Celtic” type—*i.e.*, by ponies with short-pillared molars, and only two of the eight callosities (chestnuts and ergots) invariably found on typical “forest” horses. Whether modern Shetland ponies are mainly descended from prehistoric British races or from a Norse race of the fjord type it is impossible to say. The Magdalenians, who occupied Kent’s Cavern while hunting the reindeer in the south of England

towards the close of the Old Stone Age, had no domestic animals. Neither had their successors, the Azilians, who some 8000 years ago frequented the MacArthur and other caves near Oban. The Mediterranean race (now best represented by the Basques), which followed in the wake of the Azilians, perhaps brought sheep or cattle into Britain, but there is no evidence that they possessed horses. Professor Ridgeway believes “the use of the horse by man in the British Islands cannot be placed before the end of the Bronze or the beginning of the Iron Age.”^[F17]

^[F17] ‘Origin and Influence of the Thoroughbred Horse,’ p. 92.

Through the courtesy of Sir William Turner I recently had the opportunity of examining some horse, sheep, and dog bones found near Grangemouth 30 feet below the surface, at the point where the Carron joins the Forth. Two imperfect horse skulls belonged to ponies of the “forest” type, which probably measured 12 hands at the withers, a broken sheep skull belonged to a member of the peat or turbary race, and a dog skull differed but little from that of a modern greyhound. Taking into consideration the position and nature of the deposit, one may provisionally assume that the bones belong to animals which lived in the Forth valley about the end of the Neolithic Age,—the dog and sheep undoubtedly lived under domestication, but whether the horses were tame or wild is uncertain.

The next horse bones from Scotch deposits available for study consisted of the skulls and limb-bones from Newstead, already referred to. Several of the horses in the possession of the Roman auxiliaries who garrisoned the Border Fort in the first century were over 15 hands, and had the face bent downwards on the cranium as in Kirghiz horses,—in one case the deflection is so great that the hard palate forms an angle of nearly 20° with the cranium (in the forest horse the face is in a line with the cranium). In one of the bent skulls the molars have as short pillars as in *E. stenonis* (fig. 21) of the Italian Pliocene.^[F18] Two of the Newstead skulls belong to ponies about 12 hands high, one to a pony, Arab-like in make, with metacarpals measuring 214 mm. by 28·8 mm.—i.e., to a fine-boned pony about 13 hands at the withers. The skull and teeth of the 12-hands ponies indicate that one was about two-thirds “forest,” the other two-thirds “Celtic”; the teeth as well as the skull and cannon-bones of the 13-hands pony indicate

that it was nearly a pure member of the Celtic or Libyan race. According to Dio Cassius, the Caledonians “went to war on chariots as their horses were small and fleet.” The two 12-hands Newstead ponies were probable members of the race which the Caledonians, Mætæ, and other tribes of northern Britain yoked to their war chariots. It is conceivable that soon after the Roman period ponies were taken from the mainland of Scotland to both the Western and Northern Islands. That ponies, resembling in make and size the small Newstead horses, reached Shetland some centuries before the northern islands fell into the hands of the Norsemen, is suggested by a broken pelvic bone belonging to a pony between 11 and 12 hands, found in 1911 at the Jarlshoff broch, Sumburgh, by Mr Charles M. Douglas. During the autumn of 1912, by permission of Mrs Bruce of Sumburgh, and with the help of Mr Bennet Clark, I examined a number of old hearths at Jarlshoff, probably formed about the same time as the deposit in which Mr Douglas found the broken pelvic bone.

[F18] Until the Newstead skulls were found it was believed horses with short-pillared teeth became extinct thousands of years ago.

The bones of the Celtic shorthorn and of turbary sheep, the presence of hammer-stones, pieces of pottery and scrapers, of limpet and other shells, together with the bones and implements collected during the excavations by the late Mr John Bruce of Sumburgh, support Mr Douglas’s view that horses reached Shetland some centuries before the turbulent Norse jarls, harassed by Harold Fairhair, began to settle in the Northern and Western Islands of Scotland.

It may hence be assumed that Shetland ponies are mainly descended from the “small and fleet” race yoked to the chariots of the Caledonians at the battle of Mons Graupius. This ancient race (which was probably brought to Britain during the late Celtic period) was probably originally a blend of the slender-limbed, Arab-like ponies of the Swiss Lake-dwellers and of a thick-set race of the Elephant-Bed type. That the Shetland blend is an old one is suggested by the account Herodotus gives of the horses belonging to a tribe on the north of the Danube. This tribe (the Sigynnæ), Herodotus says, “had horses with shaggy hair, five fingers long, all over their bodies, and which were small and flat-nosed, and incapable of carrying men, [F19] but when yoked under a chariot were very swift, in consequence of which the natives

drove in chariots.” Judging by this description, the chief difference between typical modern Shelties and the small horses of Central Europe in the time of Herodotus is a difference of size.

[\[F19\]](#) Herodotus probably means these small horses were incapable of carrying men into battle.

PLATE I.

Fig. 1.—A 41-inch Java pony. Many East Indian ponies are said to be saturated with Arab blood. In the Java mare figured the head and limbs bear nearly the same relation to the trunk as in small desert Arabs.

Fig. 2.—A Norwegian Udganger pony in which the legs are relatively nearly as short as in the dachshund. From a specimen in the Bergen Museum.

Fig. 3.—A 42-inch pony of the Udganger type from Iceland, in which the head and legs bear nearly the same relation to the trunk as in Exmoor ponies. This pony is characterised by a fine head, large eyes, and small ears; by fine limbs, sloping shoulders, and a short back; and by the absence of the hind chestnuts and all four ergots.

PLATE I.

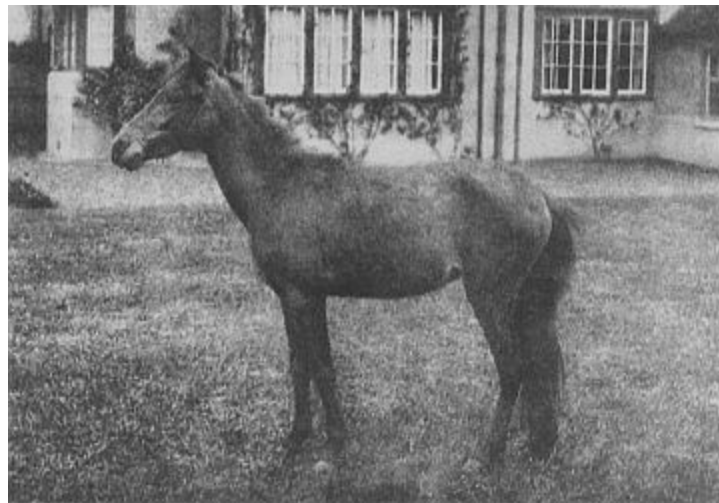


Fig. 1. A 41-INCH JAVA PONY.

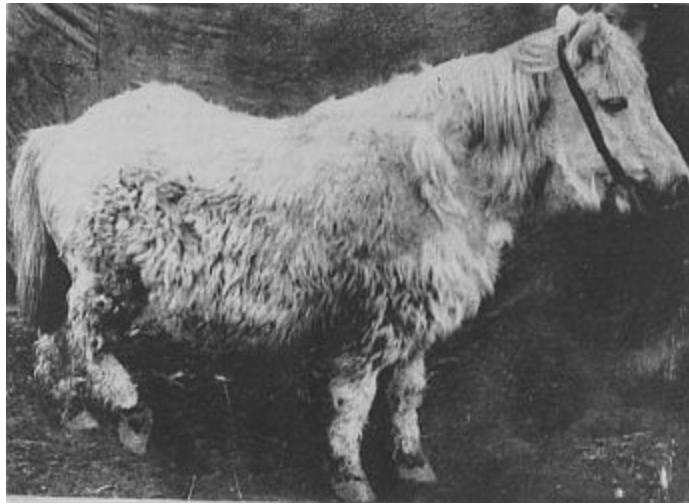


Fig. 2. A DWARFED UDGANGER PONY, NORWAY.

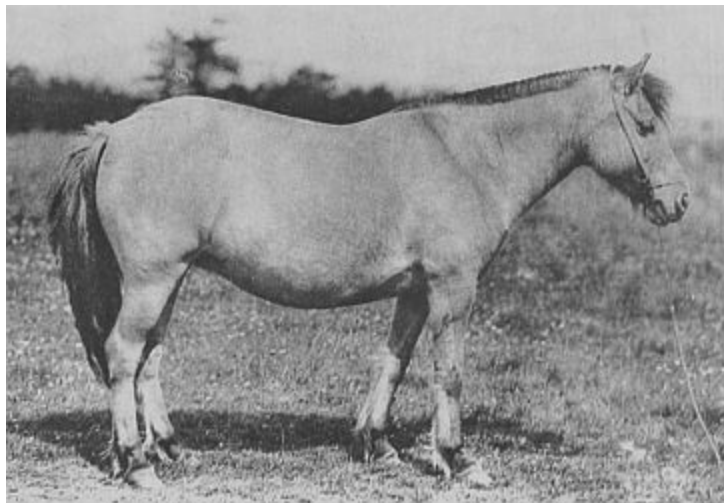


Fig. 3. A 42-INCH ICELAND PONY, UDGANGER TYPE.

PLATE II.

Fig. 4.—Skeleton of a 33-inch Shetland pony (Highland Chieftain) in the American Museum of Natural History, New York. Though the front cannon-bone is 1·5 inches shorter than in the three-toed Miocene horse *Protohippus*, it has only been dwarfed about one-quarter of an inch. From a photograph lent by Prof. H. Fairfield Osborn.

Fig. 5.—Skeleton of Persimmon, a 66-inch race-horse belonging to the late King Edward VII. The bones of the limbs of Persimmon bear practically the same relation to each other and to the trunk as in Highland Chieftain, but in Persimmon the skull is relatively shorter and the withers are relatively higher.

PLATE II.

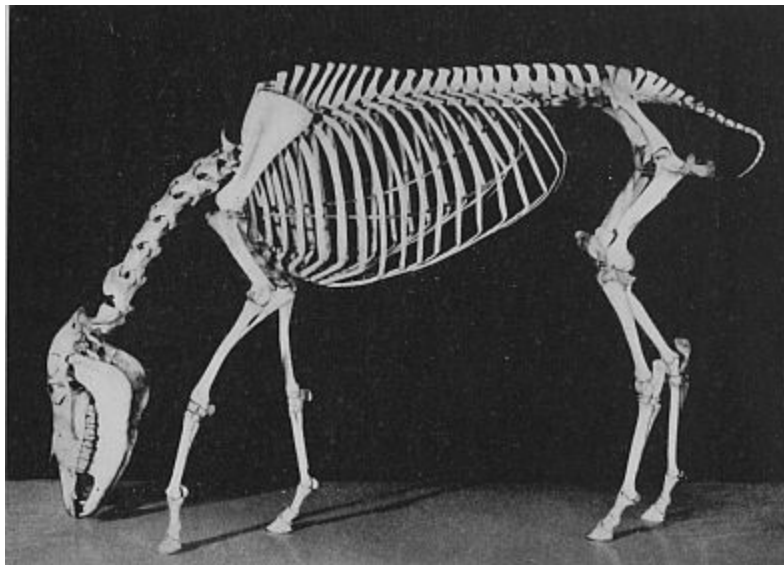


Fig. 4.—Skeleton of Highland Chieftain, a 33-inch Shetland Pony.

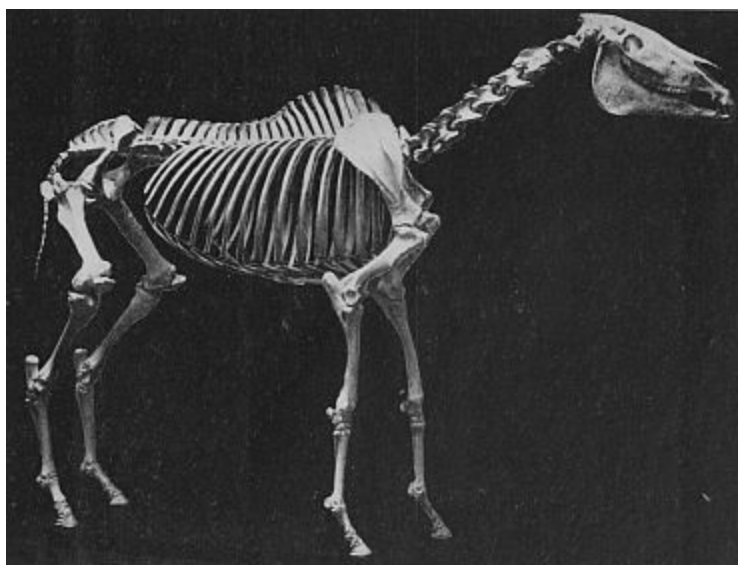


Fig. 5.—Skeleton of Persimmon, a 66-inch Thoroughbred.

PLATE III.

Fig. 6.—Side view of skull of Eric, a 36·5-inch Shetland pony of the riding type.

Fig. 7.—Skull of a new-born foal of the Celtic or riding type.

Fig. 8.—Skull of a four-year-old Prjevalsky horse from Mongolia. In the foal the cranium is relatively large and the face decidedly dished; in the Sheltie the face is longer than in the foal, but less bent downwards on the cranium and less dished; in the Prjevalsky stallion the cranium is less globular, while the face is very long and, owing to the bulging outwards of the nasals, “Roman-nosed.” Though in some Shetland ponies the face is long, the wild horse now found in Mongolia seems to have contributed little to the making of the modern Sheltie.

DIMENSIONS OF SKULLS.

	Total length. mm.	Facial length. mm.	Frontal width. mm.	Frontal index.
Prjevalsky,	518	371	187	50

Eric,	383	252	170	67
Foal at birth,	255	166	100	60
Iceland pony, “forest” type,	506	336	228	61
Pony, “Celtic” type,	494	338	185	54

PLATE III.



Fig. 6.—Skull of Eric, a 36·5-inch Shetland Pony.



Fig. 7.—Skull of a new-born foal, Celtic type.

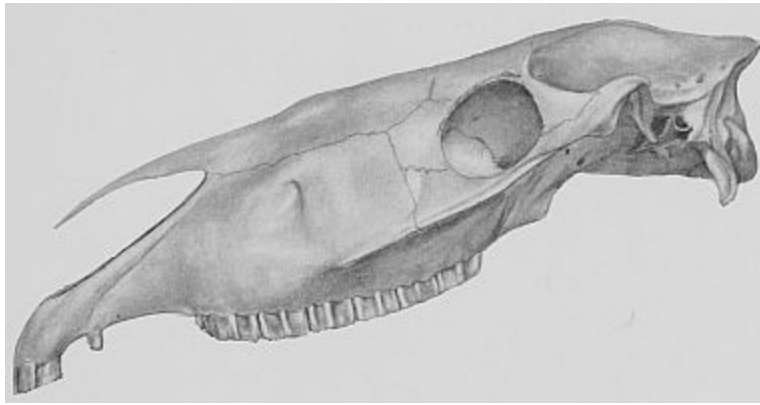


Fig. 8.—Skull of a wild Prjevalsky horse, from Mongolia.

PLATE IV.

Fig. 9a. Middle metacarpal (front cannon-bone) of Eric, a 36·5-inch Shetland pony; length, 143 mm. (5·6 inches), width at middle of shaft, 26 mm. In Eric the reduction or dwarfing of the front cannon-bones is estimated at 1 inch. Half nat. size.

Fig. 9b. Middle metacarpal of the Miocene 3-toed (36-inch?) horse *Protohippus sejunctus*; length, 177 mm. (6·9 inches), width, 21 mm.—i.e., 1·3 inches longer than in Eric. Half nat. size.

Fig. 9c. Middle metacarpal of *Hypohippus*, the (40-inch?) Miocene 3-toed “forest” horse of Montana and South Dakota; length, 215 mm. (8·4 inches), width, 22 mm.—i.e., 2·8 inches longer than in Eric. Half nat. size.

Fig. 10. Bones of fore and hind foot (half nat. size) of *Eohippus* (fig. 12). After Marsh.

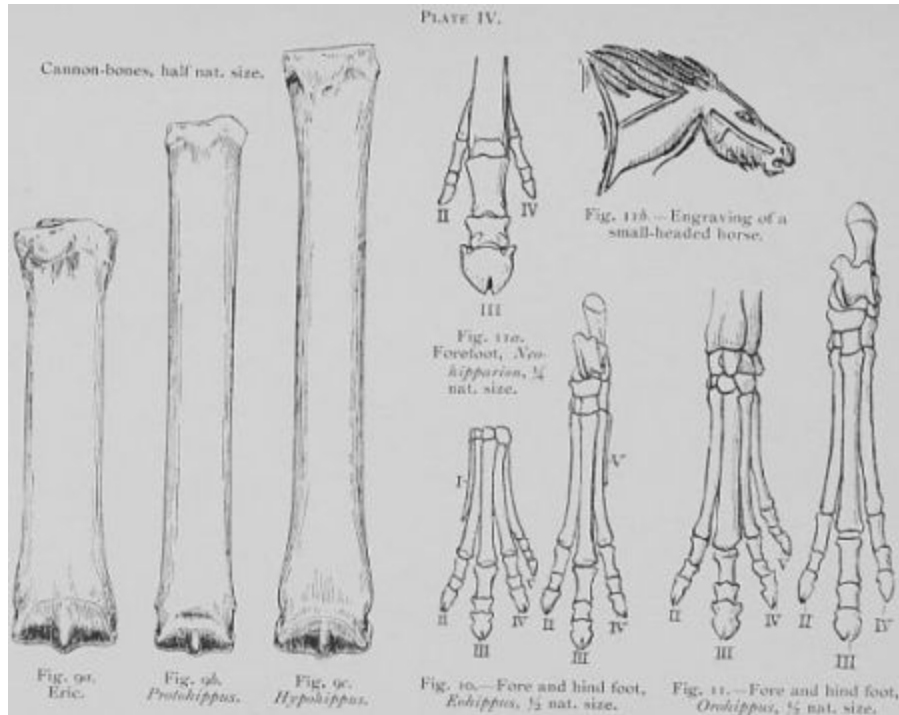
Fig. 11. Bones of fore and hind foot (half nat. size) of *Orohippus* (fig. 13). After Marsh.

Fig. 11a. Bones of the three front toes (II., III., and IV.) of *Neohipparion*, the 10-hands American Miocene desert horse with deer-like limbs. In this ancient race-horse the II. and IV. toes are very much shorter than in *Hypohippus* (fig. 20), a late Miocene “forest” horse.

Fig. 11b. Engraving of a small-headed horse made during the Early Stone Age in the Combarelles Cave, France. The short face, small ear, and

flowing mane suggest a race to which Shelties may be related. One-fourth nat. size.

PLATE IV.



Cannon-bones, half nat. size.

Fig. 9a.

Eric.

Fig. 9b.

Protohippus.

Fig. 9c.

Hypohippus.

Fig. 10.—Fore and hind foot, *Eohippus*, $\frac{1}{2}$ nat. size.

Fig. 11.—Fore and hind foot, *Orohippus*, $\frac{1}{2}$ nat. size.

Fig. 11a.

Forefoot, *Neohipparion*, $\frac{1}{4}$ nat. size.

Fig. 11b.—Engraving of a small-headed horse.

PLATE V.

Fig. 12.—Restoration of *Eohippus*, the American *Hyracotherium*, size about 12 inches. See fig. 10.

Fig. 13.—*Orohippus*, a late Eocene four-toed horse, size about 16 inches. See fig. 11.

Fig. 14.—*Mesohippus*, an Oligocene horse about 24 inches. In some of the three-toed Oligocene horses the cannon-bones were as long as in a 32-inch Shetland pony. See fig 18.

Fig. 15.—*Hypohippus*, a three-toed Miocene “forest” horse, with the II. and IV. toes long and functional. In 40-inch specimens of *Hypohippus* the cannon-bones were 2·5 inches longer than in some 40-inch Shetland ponies. See figs. 9 and 20.

Fig. 16.—*Merychippus*, a 9-hands three-toed Miocene horse. *Protohippus* (a possible ancestor of modern fine-limbed breeds) and the extinct *Hipparions* seem to have been derived from *Merychippus*.

Fig. 17.—A 33-inch Shetland pony. In modern Shelties the legs are relatively shorter than in the three-toed horses of the later Miocene deposits. Shetland ponies have probably partly sprung from ancestors allied to *Merychippus* and partly from ancestors with limbs of the *Hypohippus* type.

Figs. 12 to 16 after Osborn and Lull.

PLATE V.

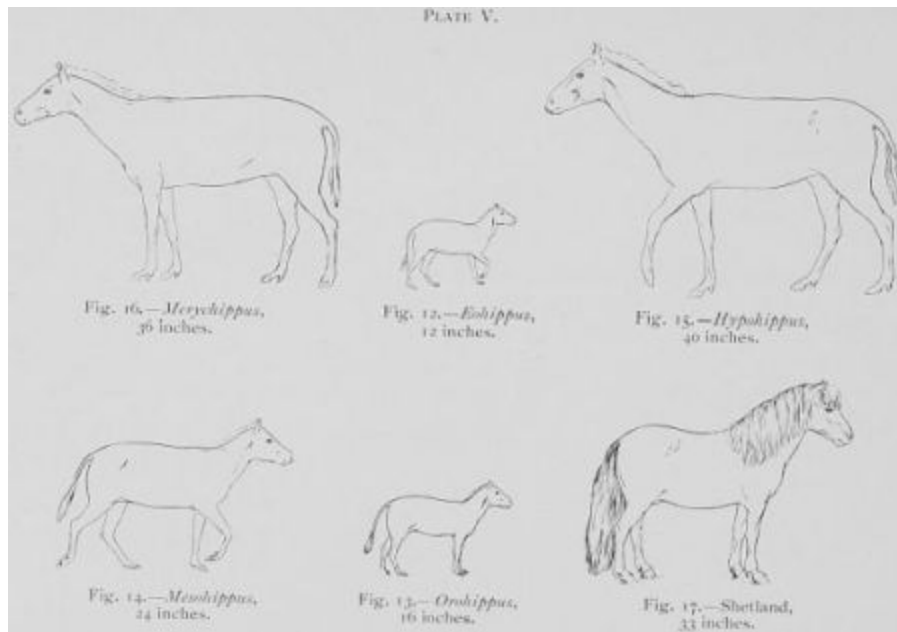


Fig. 16.—*Merychippus*, 36 inches.

Fig. 12.—*Eohippus*, 12 inches.

Fig. 15.—*Hypohippus*, 40 inches.

Fig. 14.—*Mesohippus*, 24 inches.

Fig. 13.—*Orohippus*, 16 inches.

Fig. 17.—Shetland, 33 inches.

All the figures one-thirtieth nat. size.

PLATE VI.

Fig. 18.—Skeleton of fore-foot of *Mesohippus*. The II., III., and IV. digits are complete, the V. is represented by the upper end of the metacarpal. Oligocene of America. After Marsh.

Fig. 19.—Forefoot of *Merychippus* (or *Protohippus*). Digits II. and IV. shorter than in fig. 18, and the vestige of digit V. very small or absent. American Miocene. After specimen in American Museum of Natural History.

Fig. 20.—Forefoot of *Hypohippus*, the Miocene “forest” horse. Digits II. and IV. long as in *Mesohippus*, digits I. and IV. represented by small “splints” not seen in figure. After specimen in American Museum of Natural History.

Fig. 21.—Upper molar, *E. stenonis*, natural size. The internal pillar (p) is only one-third the length of the grinding surface of the crown. Pliocene. After Boule.

Fig. 22.—Upper molar, *E. fossilis*, natural size. The internal pillar (p) is more than half the length of the grinding surface of crown. Pleistocene. Kent’s Cavern, Devonshire. After Owen.

Fig. 23.—Premolar and molars (natural size) of a small mediæval? horse from Aberdour, Aberdeenshire. The internal pillars are short. Small horses with short-pillared teeth have lived in Europe since the end of the Pliocene. In the 36·5-inch Shetland pony Eric the molars very closely agree with those figured. From specimens received from Prof. Arthur Thomson, Aberdeen.

Fig. 24.—Premolar and molars of a small horse from the Roman Fort, Newstead. The pillars are long, as in *E. robustus* of Solutrè and other small Pleistocene horses of Europe; as in the 11-hands *E. tau* of the Mexican Pleistocene; and as in Shetland ponies of the “forest” or Londonderry type.

PLATE VI.

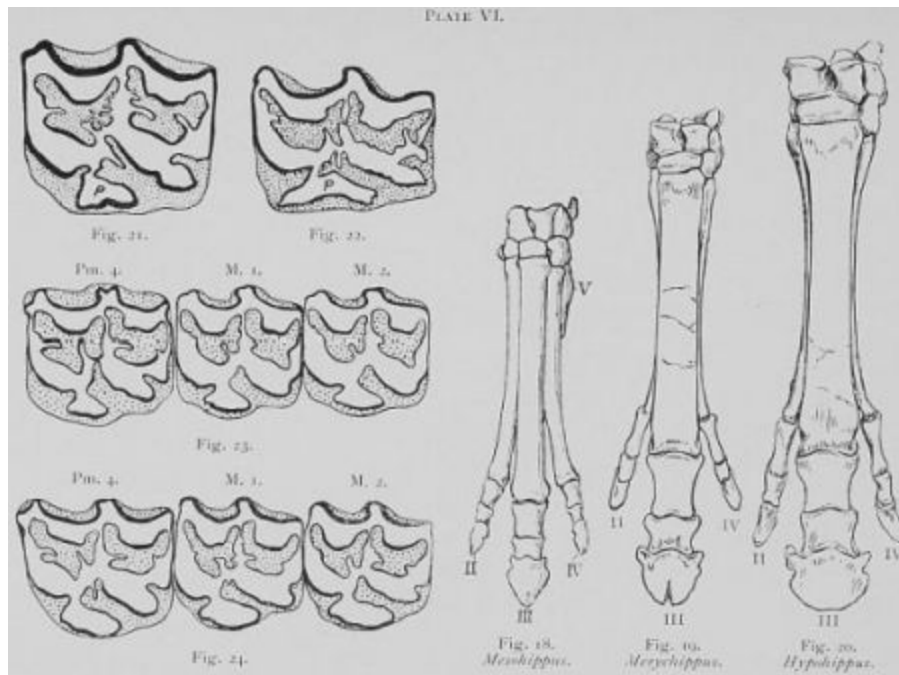


Fig. 21.

Fig. 22.

Fig. 23.

Fig. 24.

Fig. 18.

Meshippus.

Fig. 19.

Merychippus.

Fig. 20.

Hypohippus.

NOTES.

[1] Quoted in R. Holinshed, 'The Firste Volume of the Chronicles of England, Scotland, and Ireland,' 1577. Description of Scotland, pp. 21, 22; confirmed by R. Colt Hoare, 'History of Ancient Wiltshire,' Roman Era, 1812, vol. ii. p. 11.

[2] Quoted by C. Hamilton Smith, 'The Naturalist's Library,' 1873, vol. xii. p. 120. But the author does not indicate to which St Austin he refers, nor does he give any clue to the quotation, which we have been unable to verify.

[3] 'A Compendious History of the Goths, Swedes, and Vandals and other Northern Nations,' written by Olaus Magnus, Archbishop of Upsala and Metropolitan of Sweden, Eng. trans., 1658, Book II., p. 28.

[4] Olaus Magnus, *loc. cit.*, Book XXIII., p. 174.

[5] Gervase Markham, 'Cavalarice or the English Horseman,' 1607, Book I., p. 16.

[6] G. Vigfússon and F. York Powell, 'Corpus Poeticum Boreale,' 1883. W. Wagner, 'Asgard and the Gods,' trans. M. W. Macdowall, 2nd ed., 1882.

[7] R. Tudor, 'The Orkneys and Shetlands,' 1883, pp. 46, 47.

[8] 'Orkneyinga Saga,' trans. J. A. Hjalatalin and G. Goudie, 1873, p. 150.

[9] Paz Salas, 'La Felicissima Armada,' Lisbon, 1588.

[10] 'Certeine Advertisements out of Ireland,' 1588. Collected tracts on the Armada, British Museum.

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[12] J. Cossar Ewart, "The Multiple Origin of Horses and Ponies;" 'Transactions of the Highland and Agricultural Society,' 1904.

[13] W. Ridgeway, 'The Origin and Influence of the Thoroughbred Horse,' 1905.

[14] Ubaldini, 'Descrittioni del Regno di Scotia,' 1568, Eng. trans., 1829, p. 63; Edinburgh Bannatyne Club.

[15] "Descriptio Insularum Orchadiarum per me Jo Ben, ibidem coletem in Anno 1529." Macfarlane's 'Geographical Collection,' 1908, vol. iii. p. 304; Scottish History Society.

[16] "Oppressions of the 16th Century in the Islands of Orkney and Shetland, from original Documents;" Maitland Club Miscellany, 1859, p. 68.

[17] Matthew Mackaile, "Short relation of the most considerable things in Orkney," 1614; G. Barry, 'History of Orkney,' 1808, vol. xi. p. 456.

[18] "Acts and Statutes of the Lawting Sheriff and Justice Courts of Orkney and Zetland," 1617; Maitland Club Miscellany, 1840, p. 69.

[19] *Ibid.*, p. 69.

[20] "A Description of the Islands of Shetland, &c., by Captain John Smith, who was employed there by the Earle of Pembrock in the year 1633, and stayed a whole Twelve Month there;" Scottish History Society, 1908, p. 65.

[21] 'A General Geographical Description of Zetland,' by Hugh Leigh, minister of the Gospel in Brassie and Burs, through John Marr; no date—probable c. 1670; Scottish History Society, 1908, p. 250.

[22] Thomas Kirke, 'An account of a Tour in Scotland,' 1677; edited by P. Hume Brown, 1842, p. 32.

[23] J. Wallace, 'A Description of the Isles of Orkney,' 1693, ed. 1883, p. 16.

[24] J. Brand, 'A Brief Description of Orkney, Zetland, Pightland Firth, and Caithness;' Edinburgh, 1701, pp. 77–79. Brand was one of the ministers sent as a commission in 1700 by the General Assembly "to visit and order the Churches there."

[25] 'A Description of the Western Islands of Scotland.' To which is added 'A Brief Description of the Isles of Orkney and Shetland,' by Wm. Martin, Gent, 1703, p. 377. Martin's statement is perilously like a copy of Brand's, but he certainly did visit Shetland.

[26] 'Shetland Pony Stud-Book,' vol. i. p. xl.

[27] *E.g.*, T. Gifford, 'Historical Description of Zetland,' 1733, pp. 22, 23, 26, 98. D. Edmonstone, 'A View of the Ancient and Present State of the Zetland Isles,' Edinburgh, 1809, vol. i. p. 226; vol. ii. p. 206. G. Sinclair, 'A General View of the Agriculture of the Central Highlands of Scotland' (Shetland Isles), 1794, p. 247.

[28] 'An Exact and Authentic Account of the greatest White Herring Fishery in Scotland, carried on yearly in the Island of Zetland by the Dutch only.' To which is prefixed 'A Description of the Island by a Gentleman who resided Five Years on the Island,' London, 1750. Tracts on Orkney and Shetland, 1750–1801, p. 8; Library of Society of Antiquaries of Scotland.

[29] "Observations on the Islands of Shetland," 1801; Highland Society of Scotland's publications, vol. ii. p. 7.

[30] T. Gifford, 'Historical Description of the Zetland Islands,' 1733, ed. 1879, p. 22.

[31] Highland Society Report, *loc. cit.*

[32] 'Statistical Account of Shetland,' 1841, Section "Unst," by John and James Ingram, p. 45.

[33] 'The Rider and Driver': New York, December 1899.

[34] James Mill's Diary, 1740–1803, ed. 1889; Scottish History Society, vol. v. p. 86.

[35] H. H. Dixon ("The Druid"), 'Field and Fern,' 1865, pp. 29, 30.

[36] H. H. Dixon, *loc. cit.*, pp. 12–15.

[37] H. H. Dixon, *loc. cit.*, p. 12.

[38] James Fea, 'Considerations on the Fisheries in the Scotch Islands,' 1787, p. 86.

[39] A. Edmonstone, *loc. cit.*, vol. ii. p. 42.

[40] S. Hibbert, 'A Description of the Shetland Islands' (date?), ed. 1891, p. 157.

[41] S. Hibbert, *loc. cit.*, pp. 179, 180.

[42] R. Cowie, 'Shetland and its Inhabitants,' ed. 1874, p. 181.

[43] Gifford, *loc. cit.*, pp. 76, 77.

[44] H. H. Dixon, *loc. cit.*, p. 37.

[45] Cowie, *loc. cit.*, p. 191.

[46] 'Encyclopædia of Agriculture,' edited by Green and Young, p. 497; Article on Strangles.

[47] *Ibid.*, p. 189; Article on Parasites.

[48] See Mr Brydon's admirable Introductory Article in the 'Shetland Pony Study-Book,' vol. i. pp. xxxvii-xli.

[49] *Ibid.*

[50] Sheriff and Stewart Court Books of Zetland, 1749, at Lerwick.

[51] In A. Z.'s collection of documents relating to Shetland, in the Library of the Society of Antiquaries of Scotland.

[52] R. Tudor, 'The Orkneys and Shetland,' 1883, p. 125, a quotation from Campbell's 'Great White Fishery,' 2nd ed., 1753.

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Transcriber's Notes

The changes are as follows:

[Page 3](#)—Heards changed to Herds.

[Page 4](#)—ROCK DRAWING changed to ROCK-DRAWING.

[Page 5](#)—present day changed to present-day.

[Page 55](#)—pony-breeders changed to pony breeders.

[Page 63](#)—pit pony changed to pit-pony.

[Page 104](#)—they'l changed to they'll.

[Page 104](#)—Proprietors of Punch. changed to Proprietors of 'Punch.'

[Page 124](#)—one quarter changed to one-quarter.

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